Parametric Feedback Control for Shaping Stationary Probability Density Using Physics-Informed Neural Networks

Wantao Jia[a,b], Xianzheng Li[a,b], Xiaotong Feng[a,b,*]

[a]*School of Mathematics and Statistics, Northwestern Polytechnical University, Xi'an, 710072, China*
[b]*MOE Key Laboratory for Complexity Science in Aerospace, Northwestern Polytechnical University, Xi'an, 710072, China*

Abstract

This paper develops a physics-informed neural network framework for feedback control of the stationary probability density of stochastic dynamical systems. The objective is to obtain approximate density-matching feedback controls for prescribed stationary target densities under the Fokker–Planck–Kolmogorov constraints. It is assumed that the considered closed-loop systems admit suitable stationary distributions at fixed target-parameter values. The framework uses separate density and control networks to construct a parameter-conditioned approximation, and compares three FPK-residual treatment strategies under a common base training configuration and strategy-specific hyperparameter settings. The density tracking problem is formulated as a constrained optimization problem, and three constraint handling strategies are investigated: the soft constraint method, penalty method, and barrier method. The neural networks take both state variables and target distribution parameters as inputs, enabling parametric predictions of stationary probability density functions (PDFs) and their associated feedback controls within prescribed parameter ranges without retraining. Under the adopted training and simulation settings, the barrier method achieves the lowest reported closed-loop Monte Carlo (MC) L_1 density errors for both Single-degree-of-freedom (SDOF) targets and both levitation targets, demonstrating an accuracy advantage in these two-dimensional cases. The method rankings vary across the Duffing marginals, while the first SDOF benchmark shows a higher training cost for the barrier method. Independent MC simulations are further conducted by substituting the learned feedback controls into the original stochastic systems. Under the simulated initial conditions and numerical settings, the long-time MC response densities reproduce the principal distributional characteristics of the prescribed targets at representative parameter values. These results do not establish uniqueness of the stationary distribution or convergence from arbitrary initial distributions. Overall, the proposed framework provides a unified approach for parametric stationary probability density control without requiring a prescribed analytical feedback law, and the numerical results demonstrate its capabilities in parametric approximation, constraint handling, and closed-loop response prediction.

Keywords: Physics-informed neural networks, Stochastic dynamical systems, Fokker-Planck-Kolmogorov equation, Parametric Feedback control, Probability density tracking, Deep learning

1. Introduction

In many scientific and industrial applications, it is necessary to regulate the dynamical behavior of a system in order to meet performance and safety requirements. For stochastic dynamical systems[1, 2], traditional control objectives are often formulated in terms of statistical moments of the system response, such as the mean, variance, and covariance[3, 4]. These low-order statistics provide useful information but cannot fully characterize the stochastic response, particularly when the response exhibits non-Gaussian features, multi-modal distributions, or pronounced tail behavior [5, 6]. This limitation motivates stochastic tracking control, in which the stationary response probability density is taken as the control objective. Compared with moment control, density tracking control is more challenging because the control law must shape an entire distribution while preserving the feasibility requirements of a probability density function (PDF).

*Corresponding author
Email address: xtfeng@mail.nwpu.edu.cn (Xiaotong Feng)

The concept of PDF shaping control has attracted considerable attention in the control community over the past two decades. Early work by Wang [7] established a B-spline-based framework for controlling the output PDF in discrete time systems, which was subsequently extended to multivariate systems [8] and bounded control scenarios [9, 10]. For continuous time stochastic systems governed by stochastic differential equations (SDEs), the stationary PDF is characterized by the Fokker-Planck-Kolmogorov (FPK) equation [11, 12]. Annunziato and Borzì [13] formulated the PDF control problem as a bilinear optimal control problem governed by the FPK equation. On the analytical side, Zhu et al. [14, 15] designed tracking control for multi-degree-of-freedom nonlinear systems by exploiting exact stationary solutions, while Yang et al. [16] developed a Hamiltonian-based feedback control strategy using polynomial expansions and stochastic averaging. Stochastic averaging techniques have also been extended to systems under combined Gaussian and Poisson white noise excitations [17, 18], broadening the applicability of analytical methods to non-Gaussian stochastic environments. Alongside classical formulations, probabilistic target tracking has recently been investigated for practical systems under diffusion and jump disturbances [19], while stationary PDF tracking control has been developed for stochastic levitation systems [20].

These developments motivate consideration of analytical applicability, numerical dimensionality, and reuse across target-parameter values. First, analytical approaches that yield closed-form control expressions rely on restrictive solvability conditions and are therefore applicable only to systems with specific structures [21–23]. Second, numerical methods for the FPK equation, such as finite element methods[24, 25], finite difference methods[26], spectral methods [27], and path integral methods [28–30], can alleviate the limitations imposed by analytical solvability conditions to some extent. However, they suffer from the curse of dimensionality [31] and typically require substantial computational effort for each parameter configuration. Third, many applications require feedback controls for families of prescribed target densities with varying widths, locations, or shapes. When a parameterized analytical control expression is available, it can be evaluated directly at different parameter values, as illustrated by Eq.(25). For problems addressed through numerical approximation, a parameter-conditioned model offers a reusable mapping from target-distribution parameters to stationary PDFs and associated feedback controls. Once trained, it can be evaluated within the prescribed parameter range without separate training or a new numerical solution for each target-parameter value.

In recent years, deep learning methods have shown promising potential for solving the FPK equation and approximating response PDFs under prescribed dynamics in stochastic dynamical systems [32–35]. In particular, Physics-Informed Neural Networks (PINNs) [36–41] provide a framework for solving forward and inverse problems by integrating physical constraints into deep learning. Through automatic differentiation, PINNs incorporate governing equation and boundary condition residuals into the training objective to approximate solutions to the underlying problems. Building on these developments, stationary PDF tracking control can be formulated as an inverse design problem, in which the feedback control is sought so that the controlled stationary density approaches a prescribed target under the stationary FPK equation constraint. Related PINN-based studies provide a methodological basis for this formulation. Mowlavi and Nabi [42] investigated partial differential equation(PDE) constrained optimal control using PINNs, while Lu et al. [43] developed PINNs with hard constraints for inverse design, including PDE constrained topology optimization. In addition, neural network based equation discovery has explored constrained optimization as an alternative to unconstrained formulations [44], highlighting the importance of constraint treatment in physics-informed learning. Although these studies address different optimization objectives and inverse problems, they provide useful insights into incorporating physical constraints into neural network based inverse design for stationary density tracking control.

More directly related to this problem, Jia et al. [45] developed a neural network based approach for stationary PDF tracking control by incorporating conditions associated with asymptotic stability in distribution, the stationary FPK, detailed balance, and controller uniqueness into a weighted training objective. Their multi-output network approximates the reversible and irreversible components of the control, with the prescribed target density directly incorporated into the physical residuals. The proposed framework takes state variables as network inputs and considers fixed target distributions in its numerical examples, without demonstrating the use of a single trained model for varying target-distribution parameters within a prescribed range. Based on these developments, the present study focuses on two questions: how to reuse trained models as target-distribution parameters vary, and how different treatments of the governing-equation residual affect stationary density approximation and closed-loop stationary density matching.

The main contributions of this paper are as follows:

1. We construct a parameter-conditioned approximation of stationary density–control pairs by including state variables and target-distribution parameters as inputs to separate density and control networks. The displayed parameter sweeps assess reuse of the trained networks without separate training for each target parameter value.
2. We compare a fixed-weight soft-constraint formulation, an FPK-residual penalty formulation, and an auxiliary-variable formulation with multipliers and a clipped logarithmic term under a common base training configuration and strategy-specific hyperparameter settings. Evaluation includes stationary-density comparisons and parameter sweeps for a single-degree-of-freedom oscillator, an electromagnetic levitation system, and a four-dimensional coupled Duffing system, together with closed-loop Monte Carlo(MC) validation at representative parameter values.

The remainder of this paper is organized as follows. Section 2 formulates the feedback control problem for tracking a prescribed stationary PDF and presents the constrained optimization formulation. Section 3 presents the proposed PINNs-based strategy together with the constraint handling techniques used for training. Section 4 demonstrates the performance of the proposed methods through numerical examples. Section 5 concludes the paper.

2. Feedback control for targeting a prescribed stationary PDF

2.1. Problem setup

Consider a controlled nonlinear stochastic system described by the Itô SDE

$$\mathrm{d}X_i = f_i(\mathbf{X},\boldsymbol{\theta})\mathrm{d}t + u_i(\mathbf{X},\boldsymbol{\theta})\mathrm{d}t + \sum_{k=1}^{m}\sigma_{ik}(\mathbf{X},\boldsymbol{\theta})\mathrm{d}B_k(t), \quad i=1,2,\ldots,n, \tag{1}$$

where $\mathbf{X} = (X_1,\ldots,X_n)^{\mathrm{T}} \in \mathbb{R}^n$ is the state vector, $f_i : \mathbb{R}^n \times \mathbb{R}^q \to \mathbb{R}$ are the drift coefficients representing the uncontrolled dynamics, $\sigma_{ik} : \mathbb{R}^n \times \mathbb{R}^q \to \mathbb{R}$ are the diffusion coefficients, $u_i : \mathbb{R}^n \times \mathbb{R}^q \to \mathbb{R}$ are the feedback control forces to be designed, and $\{B_k(t)\}_{k=1}^m$ are independent standard Brownian motions. The parameter vector $\boldsymbol{\theta} \in \Omega_{\boldsymbol{\theta}} \subset \mathbb{R}^q$ collects the target-distribution parameters considered in the parametric study, such as parameters controlling the width, location, or shape of the prescribed density.

The stationary PDF of the system Eq.(1), if it exists, is governed by the FPK equation

$$\mathcal{L}[p,u;\boldsymbol{\theta}] \coloneqq -\sum_{i=1}^{n}\frac{\partial}{\partial x_i}\left[a_i(\mathbf{x},\boldsymbol{\theta})p(\mathbf{x},\boldsymbol{\theta})\right] + \frac{1}{2}\sum_{i,j=1}^{n}\frac{\partial^2}{\partial x_i \partial x_j}\left[b_{ij}(\mathbf{x},\boldsymbol{\theta})p(\mathbf{x},\boldsymbol{\theta})\right] = 0, \tag{2}$$

where $\mathbf{x} \in \mathbb{R}^n$ denotes the state space variable, $p(\mathbf{x},\boldsymbol{\theta})$ is the stationary PDF, and the coefficients are defined as

$$a_i(\mathbf{x},\boldsymbol{\theta}) = f_i(\mathbf{x},\boldsymbol{\theta}) + u_i(\mathbf{x},\boldsymbol{\theta}), \quad b_{ij}(\mathbf{x},\boldsymbol{\theta}) = \sum_{k=1}^{m}\sigma_{ik}(\mathbf{x},\boldsymbol{\theta})\sigma_{jk}(\mathbf{x},\boldsymbol{\theta}). \tag{3}$$

The stationary PDF must satisfy the boundary condition

$$\lim_{|\mathbf{x}|\to+\infty} p(\mathbf{x},\boldsymbol{\theta}) = 0, \tag{4}$$

and the normalization condition

$$\int_{\mathbb{R}^n} p(\mathbf{x},\boldsymbol{\theta})\,\mathrm{d}\mathbf{x} = 1. \tag{5}$$

Additionally, as a probability density, $p(\mathbf{x},\boldsymbol{\theta})$ must be non-negative:

$$p(\mathbf{x},\boldsymbol{\theta}) \geq 0, \quad \forall\, \mathbf{x} \in \mathbb{R}^n. \tag{6}$$

The control objective is to design the feedback control $\mathbf{u}(\mathbf{x},\boldsymbol{\theta})$ such that the stationary PDF of the controlled system matches a prescribed target PDF $p_{\mathrm{target}}(\mathbf{x},\boldsymbol{\theta})$. Here, the target PDF is parameterized by $\boldsymbol{\theta}$, which allows for

a unified treatment in which a single pair of neural networks can be evaluated over the prescribed target-parameter range without parameter-specific retraining.

For each fixed $\boldsymbol{\theta}$, this study concerns stationary density shaping rather than transient distribution tracking. The interpretation of p as a closed-loop stationary PDF presupposes that the considered controlled system admits a suitable stationary distribution with a density satisfying the stated integrability and boundary requirements. A density–control pair that approximately satisfies the stationary FPK equation does not by itself establish uniqueness of that stationary distribution or convergence to it from arbitrary initial distributions. The parameterized networks represent a family of fixed-parameter stationary design problems; the target parameters are held constant in each closed-loop MC simulation.

2.2. Constrained optimization formulation

The control design problem can be formulated as a constrained optimization problem. The objective is to minimize the discrepancy between the stationary PDF $p(\mathbf{x}, \boldsymbol{\theta})$ of the controlled system and the target PDF $p_{\text{target}}(\mathbf{x}, \boldsymbol{\theta})$. Following the performance index formulations in [8, 16], we define the cost functional as the L^2 distance between the actual and target PDFs:

$$\mathcal{J}[p; \boldsymbol{\theta}] = \int_{\mathbb{R}^n} \left[p(\mathbf{x}, \boldsymbol{\theta}) - p_{\text{target}}(\mathbf{x}, \boldsymbol{\theta})\right]^2 \mathrm{d}\mathbf{x}. \tag{7}$$

For a prescribed feedback control $\mathbf{u}$ that admits a suitable stationary density, let p denote a corresponding stationary PDF; uniqueness is not assumed. Minimizing $\mathcal{J}[p; \boldsymbol{\theta}]$ over admissible density–control pairs seeks to reduce the stationary PDF discrepancy in the L^2 sense. This objective contains no control-energy term, input-amplitude penalty, or transient-response cost, so optimality refers only to the stated stationary density-matching objective and does not imply optimal overall control performance. The finite-domain objective in Eq.(9a) is its numerical counterpart. The stationary density-matching design problem over $\boldsymbol{\theta} \in \Omega_{\boldsymbol{\theta}}$ is formulated as:

$$\min_{\mathbf{u}(\mathbf{x}, \boldsymbol{\theta})} \quad \mathcal{J}[p; \boldsymbol{\theta}], \quad \forall\, \boldsymbol{\theta} \in \Omega_{\boldsymbol{\theta}}, \tag{8a}$$

$$\text{s.t.} \quad \mathcal{L}[p, u; \boldsymbol{\theta}] = 0, \tag{8b}$$

$$p(\mathbf{x}, \boldsymbol{\theta}) \geq 0, \tag{8c}$$

$$\lim_{|\mathbf{x}| \to +\infty} p(\mathbf{x}, \boldsymbol{\theta}) = 0, \tag{8d}$$

$$\int_{\mathbb{R}^n} p(\mathbf{x}, \boldsymbol{\theta})\, \mathrm{d}\mathbf{x} = 1. \tag{8e}$$

Because the same stationary target PDF may be induced by more than one feedback control, the inverse control problem is not necessarily unique. The numerical result should be interpreted as an approximate density-matching feedback control obtained under the adopted neural-network parameterization and optimization procedure, rather than as a certified globally optimal control.

2.3. Finite-domain approximation

The boundary condition Eq.(8d) and normalization condition Eq.(8e) are defined over the infinite domain $\mathbb{R}^n$, which poses computational challenges. Following standard practice in numerical methods for the FPK equation [26, 33], we approximate the problem on a finite computational domain $\Omega_{\mathbf{x}} \subset \mathbb{R}^n$ that is chosen to be sufficiently large to capture the bulk of the probability mass. The boundary condition Eq.(8d) is then replaced by a homogeneous Dirichlet condition on the boundary $\partial\Omega_{\mathbf{x}}$.

The approximated constrained optimization problem becomes:

$$\min_{\mathbf{u}(\mathbf{x},\boldsymbol{\theta})} \quad \mathcal{J} = \int_{\Omega_{\mathbf{x}}} \left[p(\mathbf{x},\boldsymbol{\theta}) - p_{\text{target}}(\mathbf{x},\boldsymbol{\theta})\right]^2 \mathrm{d}\mathbf{x}, \quad \forall\, \boldsymbol{\theta} \in \Omega_{\boldsymbol{\theta}}, \tag{9a}$$

$$\text{s.t.} \quad \mathcal{L}[p, u; \boldsymbol{\theta}] = 0, \quad (\mathbf{x},\boldsymbol{\theta}) \in \Omega_{\mathbf{x}} \times \Omega_{\boldsymbol{\theta}}, \tag{9b}$$

$$p(\mathbf{x},\boldsymbol{\theta}) \geq 0, \quad (\mathbf{x},\boldsymbol{\theta}) \in \Omega_{\mathbf{x}} \times \Omega_{\boldsymbol{\theta}}, \tag{9c}$$

$$p(\mathbf{x},\boldsymbol{\theta}) = 0, \quad (\mathbf{x},\boldsymbol{\theta}) \in \partial\Omega_{\mathbf{x}} \times \Omega_{\boldsymbol{\theta}}, \tag{9d}$$

$$\int_{\Omega_{\mathbf{x}}} p(\mathbf{x},\boldsymbol{\theta})\, \mathrm{d}\mathbf{x} = 1, \quad \boldsymbol{\theta} \in \Omega_{\boldsymbol{\theta}}. \tag{9e}$$

Remark 1. The optimization problem Eq.(9) seeks a control mapping $\mathbf{u}(\mathbf{x},\boldsymbol{\theta})$ over the prescribed target-parameter range. Its neural approximation is trained jointly across sampled parameter values, rather than separately for each fixed target.

Remark 2. The non-negativity of the learned PDF is imposed for all three training strategies through the exponential output transformation of the PDF network. The soft, penalty, and barrier formulations considered in Section 3 therefore differ in how the FPK residual is treated: the soft formulation uses a fixed residual weight, the penalty formulation progressively increases the residual weight, and the barrier formulation combines an auxiliary relation with a clipped logarithmic term. The barrier term is unrelated to the non-negativity of the PDF.

3. The physics-informed neural network framework

In this section, we present a physics-informed neural network (PINN) framework to solve the constrained optimization problem Eq.(9). The key idea is to parameterize both the stationary PDF $p(\mathbf{x},\boldsymbol{\theta})$ and the control force $\mathbf{u}(\mathbf{x},\boldsymbol{\theta})$ using deep neural networks, and to train these networks by minimizing a loss function that encodes the objective Eq.(9a) and the constraints Eq.(9b)–Eq.(9e). The original control problem is formulated as a constrained optimization problem. In the PINN implementation, we solve a sampled unconstrained surrogate in which the FPK equation, boundary condition, and normalization condition are incorporated through residual-based loss terms. Therefore, the learned solution should be interpreted as an approximate solution under the adopted neural-network parameterization and sampling strategy.

3.1. Neural network architecture

The prescribed target PDF $p_{\text{target}}(\mathbf{x},\boldsymbol{\theta})$ specifies the desired distribution, whereas the stationary PDF $p(\mathbf{x},\boldsymbol{\theta})$ represents the response induced by the feedback control. These two densities need not coincide, since an arbitrary target may not be exactly attainable under the given dynamics and control structure. We therefore approximate the response PDF and the feedback control with separate networks and optimize them jointly under the tracking objective and the FPK, boundary, and normalization constraints. This formulation allows the learned response to balance target tracking and dynamical consistency without assuming exact target attainability. For an analytically specified attainable target, fixing the PDF and solving only for the control is also a valid alternative. We do not claim that the two-network formulation is universally more accurate or computationally preferable. Moreover, small sampled loss values alone do not establish that the learned PDF equals the stationary PDF induced by the learned control; this consistency is further assessed through closed-loop MC simulations under the simulated initial conditions and numerical settings. The validation tasks and their scope are summarized in Section 4.

To address the parametric control problem, we employ two fully connected deep neural networks. The first network, denoted by $p_{\text{NN}}(\mathbf{x},\boldsymbol{\theta};\boldsymbol{\omega}_p)$, approximates the stationary PDF $p(\mathbf{x},\boldsymbol{\theta})$, where $\boldsymbol{\omega}_p$ represents the trainable parameters (weights and biases) of this network. The second network, denoted by $\mathbf{u}_{\text{NN}}(\mathbf{x},\boldsymbol{\theta};\boldsymbol{\omega}_u)$, approximates the control force $\mathbf{u}(\mathbf{x},\boldsymbol{\theta})$, with trainable parameters $\boldsymbol{\omega}_u$. Both networks take as input the concatenated vector $[\mathbf{x},\boldsymbol{\theta}]$, enabling the networks to produce outputs that depend on both the state variables and the target-distribution parameters.

By including the target-distribution parameters $\boldsymbol{\theta}$ as network inputs, the trained models can be evaluated at different parameter values without separate training for each target. The displayed parameter sweeps assess this reuse.

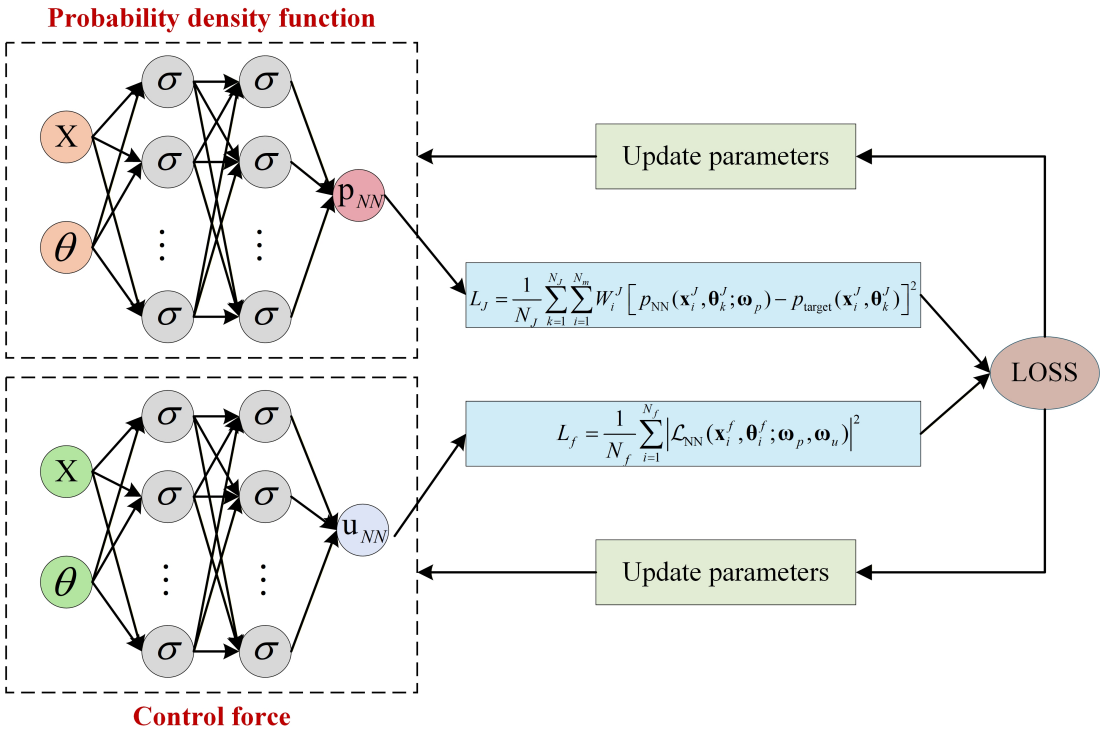


Fig. 1: Schematic of the neural-network architecture used to approximate the stationary PDF and the feedback control force.

3.2. Physics-informed loss function and training strategies

The two neural networks $p_{\mathrm{NN}}(\mathbf{x},\boldsymbol{\theta};\boldsymbol{\omega}_p)$ and $\mathbf{u}_{\mathrm{NN}}(\mathbf{x},\boldsymbol{\theta};\boldsymbol{\omega}_u)$ are coupled through the FPK operator. This coupling defines the proposed physics-informed neural-network model. The FPK operator can be evaluated as $\mathcal{L}_{\mathrm{NN}}(\mathbf{x},\boldsymbol{\theta};\boldsymbol{\omega}_p,\boldsymbol{\omega}_u) = \mathcal{L}[p_{\mathrm{NN}},\mathbf{u}_{\mathrm{NN}};\boldsymbol{\theta}]$ using automatic differentiation.

The training objective is to find the network parameters $\boldsymbol{\omega}_p^*$ and $\boldsymbol{\omega}_u^*$ that minimize a loss function encoding both the optimization objective and the constraints. The general form of the loss function is:

$$L(\boldsymbol{\omega}_p,\boldsymbol{\omega}_u) = L_J + w_f L_f + w_b L_b + w_n L_n, \tag{10}$$

where:

- L_J measures the tracking error between the network output and the target PDF, corresponding to the objective Eq.(9a);
- L_f enforces the FPK equation Eq.(9b);
- L_b enforces the boundary condition Eq.(9d);
- L_n enforces the normalization condition Eq.(9e);
- w_f, w_b, w_n are weight parameters that balance the different terms.

Each loss component is computed using collocation points sampled from the appropriate domains. Specifically, the FPK residual loss is:

$$L_f = \frac{1}{N_f}\sum_{i=1}^{N_f}\left|\mathcal{L}_{\mathrm{NN}}(\mathbf{x}_i^f,\boldsymbol{\theta}_i^f;\boldsymbol{\omega}_p,\boldsymbol{\omega}_u)\right|^2, \tag{11}$$

where $\left\{\mathbf{x}_i^f,\boldsymbol{\theta}_i^f\right\}_{i=1}^{N_f}$ are collocation points sampled from $\Omega_{\mathbf{x}}\times\Omega_{\boldsymbol{\theta}}$. Each state coordinate and target-distribution parameter is sampled independently from its prescribed interval using uniform random sampling; Latin hypercube sampling

is not used. The boundary condition loss is:

$$L_b = \frac{1}{N_b} \sum_{i=1}^{N_b} \left| p_{\mathrm{NN}}(\mathbf{x}_i^b, \boldsymbol{\theta}_i^b; \boldsymbol{\omega}_p) \right|^2, \tag{12}$$

where $\left\{\mathbf{x}_i^b, \boldsymbol{\theta}_i^b\right\}_{i=1}^{N_b}$ are sampled from $\partial\Omega_{\mathbf{x}} \times \Omega_{\boldsymbol{\theta}}$. For a rectangular two-dimensional domain, a random location is first sampled uniformly with respect to the total boundary length and is then mapped to one of the four edges. This produces random boundary points that are uniform with respect to boundary arclength.

The normalization condition Eq.(9e) and the tracking objective Eq.(9a) require integration over $\Omega_{\mathbf{x}}$. These integrals are evaluated by composite Gauss–Legendre quadrature. Each state interval is partitioned into the prescribed subintervals. On every subinterval, one-dimensional Gauss–Legendre nodes and weights are generated after mapping the standard rule from $[-1, 1]$ to that subinterval; the corresponding Jacobian factor is included in the mapped weights. The nodes and weights from all subintervals are then concatenated in each state direction before the tensor-product rule is formed. Thus, the reported numbers of nodes refer to the total numbers after concatenation, rather than to the numbers of nodes on a single subinterval. The two-dimensional examples use nine nodes in each state direction after concatenation, giving $9^2 = 81$ spatial quadrature points. The four-dimensional Duffing example uses four nodes in each of the four state directions after concatenation, giving $4^4 = 256$ spatial quadrature points. The same composite tensor-product construction is used for both the normalization and tracking integrals. Let $\{\mathbf{x}_i^n, W_i^n\}_{i=1}^{N_M}$ and $\{\mathbf{x}_i^J, W_i^J\}_{i=1}^{N_m}$ denote the tensor-product quadrature nodes and weights used for normalization and tracking, respectively. The normalization loss is approximated as:

$$L_n = \frac{1}{N_n} \sum_{j=1}^{N_n} \left| \sum_{i=1}^{N_M} W_i^n p_{\mathrm{NN}}(\mathbf{x}_i^n, \boldsymbol{\theta}_j^n; \boldsymbol{\omega}_p) - 1.0 \right|^2, \tag{13}$$

where $\{\boldsymbol{\theta}_j^n\}_{j=1}^{N_n}$ are the target-parameter values used to evaluate the normalization condition. The tracking error loss is:

$$L_J = \frac{1}{N_J} \sum_{k=1}^{N_J} \sum_{i=1}^{N_m} W_i^J \left[p_{\mathrm{NN}}(\mathbf{x}_i^J, \boldsymbol{\theta}_k^J; \boldsymbol{\omega}_p) - p_{\mathrm{target}}(\mathbf{x}_i^J, \boldsymbol{\theta}_k^J) \right]^2, \tag{14}$$

where $\{\boldsymbol{\theta}_k^J\}_{k=1}^{N_J}$ are the target-parameter values used to evaluate the tracking objective. The averages over sampled parameter values in Eq.(13) and Eq.(14) approximate the expectation over the prescribed target-parameter range.

The network parameters are optimized using gradient-based methods. At each iteration k, the parameters are updated as:

$$\begin{aligned} \boldsymbol{\omega}_p^{k+1} &= \boldsymbol{\omega}_p^k - \eta(k) \nabla_{\boldsymbol{\omega}_p} L(\boldsymbol{\omega}_p^k, \boldsymbol{\omega}_u^k), \\ \boldsymbol{\omega}_u^{k+1} &= \boldsymbol{\omega}_u^k - \eta(k) \nabla_{\boldsymbol{\omega}_u} L(\boldsymbol{\omega}_p^k, \boldsymbol{\omega}_u^k), \end{aligned} \tag{15}$$

where $\eta(k)$ is the learning rate at iteration k, and the gradients are computed via automatic differentiation.

For all three strategies, the PDF network uses an exponential output layer, so $p_{\mathrm{NN}}(\mathbf{x}, \boldsymbol{\theta}) > 0$ is satisfied by construction. The strategies therefore differ primarily in how the FPK residual is incorporated into the optimization objective. In particular, the clipped logarithmic term in the barrier formulation acts on separately updated auxiliary variables, rather than on p_{NN}. These formulations are detailed below.

3.2.1. Soft constraint method

The soft constraint method minimizes the total loss in Eq.(10) using fixed weights w_f, w_b, and w_n throughout training. As in the other two strategies, pointwise positivity is enforced by the exponential output layer. The main advantage of the soft formulation is its simplicity and relatively low computational cost. Its performance, however, may be sensitive to the fixed balance among the tracking, FPK-residual, boundary, and normalization terms.

3.2.2. Penalty method

In the penalty formulation, the boundary and normalization weights are fixed, whereas the FPK-residual weight is updated adaptively during Adam training. Its loss function is

$$L_{\text{penalty}}(\boldsymbol{\omega}_p, \boldsymbol{\omega}_u) = L_J + \rho_f(k) L_f + w_b L_b + w_n L_n. \tag{16}$$

Here $\rho_f(k)$ denotes the adaptive FPK-residual penalty parameter, corresponding to `lamda` in the implementation. During Adam training, the network parameters are updated to minimize L_{penalty}, whereas the penalty parameter is updated by supplying Adam with the gradient of the negative loss,

$$g_\rho^k = \left.\frac{\partial(-L_{\text{penalty}})}{\partial \rho_f}\right|_k = -L_f^k.$$

Thus, the penalty-parameter update performs Adam-based gradient ascent on the original loss with respect to ρ_f, adaptively increasing the emphasis on the FPK residual. The boundary and normalization weights remain fixed. Positivity is imposed by the exponential PDF output.

3.2.3. Barrier method

The strategy referred to here as the barrier method combines an auxiliary relation with a clipped logarithmic term. At each selected training point, let r_i denote the signed FPK residual:

$$r_i(\boldsymbol{\omega}_p, \boldsymbol{\omega}_u) = \mathcal{L}_{\text{NN}}\left(\mathbf{x}_i^{\text{bar}}, \boldsymbol{\theta}_i^{\text{bar}}; \boldsymbol{\omega}_p, \boldsymbol{\omega}_u\right). \tag{17}$$

The auxiliary variable s_i is updated separately from the network parameters and projected after its trial update according to

$$s_i \leftarrow \max\left\{s_i^{\text{trial}}, \varepsilon - r_i, s_{\min}\right\}, \qquad s_{\min} > 0. \tag{18}$$

Here ε is the prescribed reference value in the auxiliary relation, and $s_{\min}$ is a problem-dependent positive numerical floor. During Adam training, the objective is

$$L_{\text{barrier}} = L_{\text{main}} + \sum_{i=1}^{N_{\text{bar}}} \lambda_i (r_i + s_i - \varepsilon) - \mu \sum_{i=1}^{N_{\text{bar}}} \log(\max\{s_i, s_{\min}\}). \tag{19}$$

Here L_{main} denotes the main training loss, λ_i is the multiplier associated with the auxiliary relation, and $\mu > 0$ is the logarithmic parameter. The clipped logarithmic term acts on s_i, while the projection guarantees its positive lower bound and keeps the logarithmic evaluation well defined. The multiplier term connects s_i to the signed residual through $r_i + s_i - \varepsilon$.

During Adam training, the logarithmic parameter is updated by a multiplicative decay with a positive lower bound,

$$\mu_{k+1} = \max\{\beta_k \mu_k, \mu_{\min}\}, \qquad 0 < \beta_k < 1, \quad \mu_{\min} > 0,$$

where β_k is the decay factor used at update k and $\mu_{\min}$ is the prescribed lower bound. This rule corresponds to `update_mu` in the implementation.

For comparison, the three strategies use the same neural-network architectures, numerical precision, numbers and sampling rules of the collocation and boundary points, quadrature construction, optimizers, number of Adam epochs, and L-BFGS iteration limit. The common training configuration is summarized in Table 1. The differences are restricted to the strategy-specific treatment of the FPK residual: a fixed weight is used in the soft formulation, an iteration-dependent residual weight is used in the penalty formulation, and an auxiliary relation and a clipped logarithmic term are used during Adam training in the barrier formulation.

4. Numerical examples

Unless otherwise stated, all numerical examples use the training configuration summarized in Table 1. The implementation uses double precision and a two-stage optimization strategy, in which Adam is first employed for stable initial training and L-BFGS is then used for further refinement, with a maximum of 5000 iterations. For every parameterized target PDF, the normalization constant is evaluated at each value of ω. For the second Duffing target, the analytical constant given below normalizes the density over $\mathbb{R}^4$. The finite-domain normalization used for MC error comparisons is specified in Eq.(20).

The numerical results address three complementary tasks: comparison with a selected analytical control, approximation of learned PDFs at displayed state and parameter values, and closed-loop response assessment by long-time MC simulation. The first task is used for the first Single-degree-of-freedom (SDOF) target; the second uses pointwise absolute PDF errors and cross-sectional profiles; the third uses joint PDFs for the two-dimensional systems and two-dimensional marginals for the Duffing system. In each MC figure, the upper row shows estimated densities and the lower row shows pointwise absolute errors and integrated L_1 errors. For the latter, the target and MC-estimated densities are normalized over the same comparison domain Ω and denoted by $p_{t,\Omega}$ and $p_{\mathrm{MC},\Omega}$. The reported error is defined by

$$E_1 = \int_{\Omega} |p_{t,\Omega}(\mathbf{x}) - p_{\mathrm{MC},\Omega}(\mathbf{x})| \, \mathrm{d}\mathbf{x}. \tag{20}$$

Here, $\Omega = \Omega_{\mathbf{x}}$ for the two-dimensional joint-PDF comparisons; for the Duffing system, Ω is the corresponding two-dimensional marginal domain. In the grid-based implementation, the absolute density difference in each bin is weighted by the associated cell area. Thus, the reported L_1 values are density-integral errors evaluated using the same domain and normalization convention for the target and MC PDFs.

The pointwise error plots use the absolute density difference $e = |p_{\mathrm{NN}} - p_{\mathrm{target}}|$. For $e > 0$, the logarithmic error panels display $\log_{10} e$; this logarithm is undefined at $e = 0$. Very low values or sharp downward minima can reflect intersections between predicted and target density curves. The absolute-error plots and the displayed PDF cross-sectional profiles do not, by themselves, quantify relative density errors, tail-probability errors, or errors in response variance.

Table 1: Training configuration used in the numerical examples.

Item	**Setting**
Network type	Parametric PINN consisting of two neural networks, p_{NN} (`net_pdf`) and $\mathbf{u}_{\mathrm{NN}}$ (`net_control`), which are optimized jointly through the total physics-informed loss.
Data type	`float64`.
p_{NN} (`net_pdf`) architecture	Four hidden layers $[128, 128, 128, 128]$ with `tanh` activation and an exponential output layer of dimension one.
$\mathbf{u}_{\mathrm{NN}}$ (`net_control`) architecture	Three hidden layers $[64, 64, 64]$ with `tanh` activation and a linear output layer of dimension d_u. Here, $d_u = 1$ for the SDOF and levitation examples and $d_u = 2$ for the coupled Duffing example.
Interior residual points	10000 points. Each state coordinate and target-distribution parameter is sampled independently from its prescribed interval using uniform random sampling; Latin hypercube sampling is not used.
Boundary points	1000 points. For a rectangular two-dimensional domain, a boundary location is sampled uniformly with respect to the total perimeter and then mapped to one of the four edges.
Normalization quadrature	Composite Gauss–Legendre quadrature is used. Nodes and mapped weights are generated on each prescribed subinterval, concatenated in each state direction, and then combined by a tensor-product grid. The reported node counts are the totals after concatenation: nine nodes per state direction for the two-dimensional examples, giving $9^2 = 81$ spatial points; and four nodes per state direction for the four-dimensional Duffing example, giving $4^4 = 256$ spatial points. With 10 parameter points, these rules give 810 and 2560 evaluations, respectively.
Objective-function quadrature	The same composite tensor-product Gauss–Legendre construction is used. The two-dimensional examples use 81 spatial points and 20 parameter points, giving 1620 evaluations; the four-dimensional Duffing example uses 256 spatial points and 20 parameter points, giving 5120 evaluations.
Batch size	10000, corresponding to full-batch training for the residual points.
First-stage optimizer	Adam.
Adam learning rate	Piecewise learning rate with initial value 5×10^{-4}; the decay boundary is 5000, and the corresponding values are $[5 \times 10^{-4}, 3 \times 10^{-4}]$.
Adam epochs	10000 epochs.
Second-stage optimizer	L-BFGS, with a maximum of 5000 iterations.

4.1. SDOF system

Consider the following single-degree-of-freedom example

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= -x_1 + (\alpha - \beta x_1^2)x_2 + u + \xi(t) + x_1\zeta(t), \end{aligned} \tag{21}$$

where $\xi(t)$ and $\zeta(t)$ are two independent Gaussian white-noise processes with noise intensities $2D$ and $2S$, respectively, and u is the control force. The system parameters used in this example are $\alpha = -0.08$, $\beta = 0.1$, $D = 0.001$, and $S = 0.004$.

The corresponding reduced FPK equation is

$$0 = -\frac{\partial}{\partial x_1}\left[a_1 p(x_1, x_2)\right] - \frac{\partial}{\partial x_2}\left[a_2 p(x_1, x_2)\right] + \frac{1}{2}\frac{\partial^2}{\partial x_2^2}\left[b_{22} p(x_1, x_2)\right], \tag{22}$$

where

$$\begin{aligned} a_1 = x_2, \quad a_2 = -x_1 + (\alpha - \beta x_1^2)x_2 + u, \\ b_{22} = 2D + 2Sx_1^2, \quad b_{11} = b_{12} = b_{21} = 0. \end{aligned} \tag{23}$$

4.1.1. The first target PDF

The specified target stationary probability density function is

$$p_{\text{target}}(x_1, x_2) = C(\omega)\exp\left[a(x_2^2 + \omega^2 x_1^2) - b(x_2^2 + \omega^2 x_1^2)^2\right], \tag{24}$$

where $a = 1.0$ and $b = 1.5$. The computations for this target are performed on $\Omega_{\mathbf{x}} = [-1.2, 1.2] \times [-1.5, 1.5]$, with the target-distribution parameter sampled over $\omega \in [1.1, 1.3]$.

For the first SDOF target, an explicitly selected analytical reference control is used as a pointwise benchmark. For $a = 1.0$ and $b = 1.5$, the analytical feedback control is given by

$$u_{\text{exact}}(x_1, x_2; \omega) = (1 - \omega^2)x_1 + \left(2D - \alpha - 6S\omega^2 x_1^4\right)x_2 + \left(\beta - 6D\omega^2 + 2S\right)x_1^2 x_2 - 6\left(D + Sx_1^2\right)x_2^3. \tag{25}$$

This selected analytical expression provides a pointwise benchmark for evaluating the learned control force; it is not supplied as an input to training. A difference from a selected reference law does not, by itself, imply failure to induce the target density when alternative controls are admissible.

Fig. 2 compares the learned controls with the selected analytical reference at $\omega = 1.10$, 1.14, and 1.18. This provides a pointwise control-reconstruction benchmark complementary to the subsequent density comparisons; disagreement with this selected law alone does not determine the quality of an induced density when controls are non-unique.

Fig. 3 shows that the learned PDFs reproduce the low-density interior and surrounding high-probability ring of the annular target. The error maps identify deviations along the ridge and its transition regions.

Fig. 4 presents representative one-dimensional cuts through the annular target together with the corresponding pointwise absolute errors. These plots compare agreement with the target density curve at the displayed coordinates. Smaller errors on these cuts indicate closer pointwise agreement in absolute terms.

Fig. 5 evaluates one trained barrier density network at the six parameter values listed in the caption. Errors are concentrated near the annular ridge and its transitions.

Fig. 6 assesses whether the learned controls recover the annular concentration in simulated responses. The reported L_1 errors are 0.119, 0.078, and 0.0693 for the soft, penalty, and barrier formulations, respectively.

Table 2: Computational time of the three constraint-handling methods.

Method	Soft method	Penalty method	Barrier method
Time(s)	12019.44	15738.33	25746.60

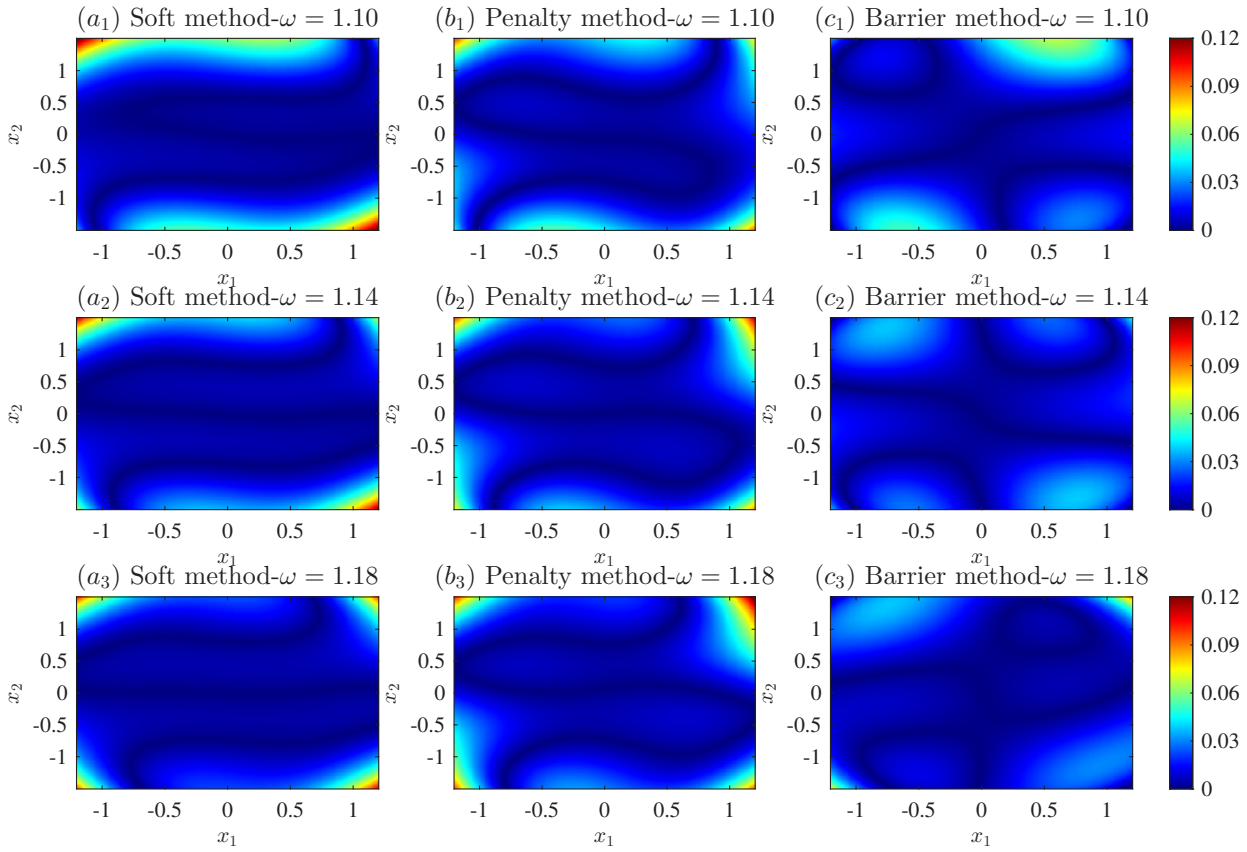


Fig. 2: Pointwise absolute errors between the learned control forces and the analytical reference control for the first SDOF target PDF at $\omega = 1.10$, 1.14, and 1.18.

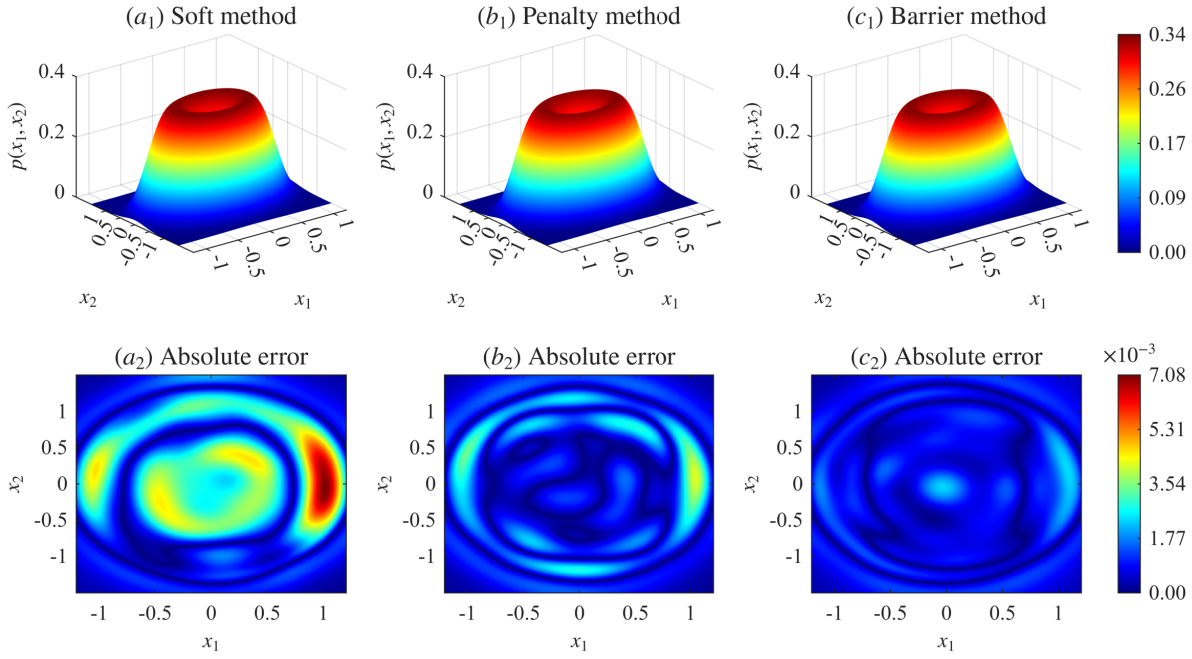


Fig. 3: Comparison of learned stationary PDFs and their pointwise absolute errors for the first SDOF target PDF.

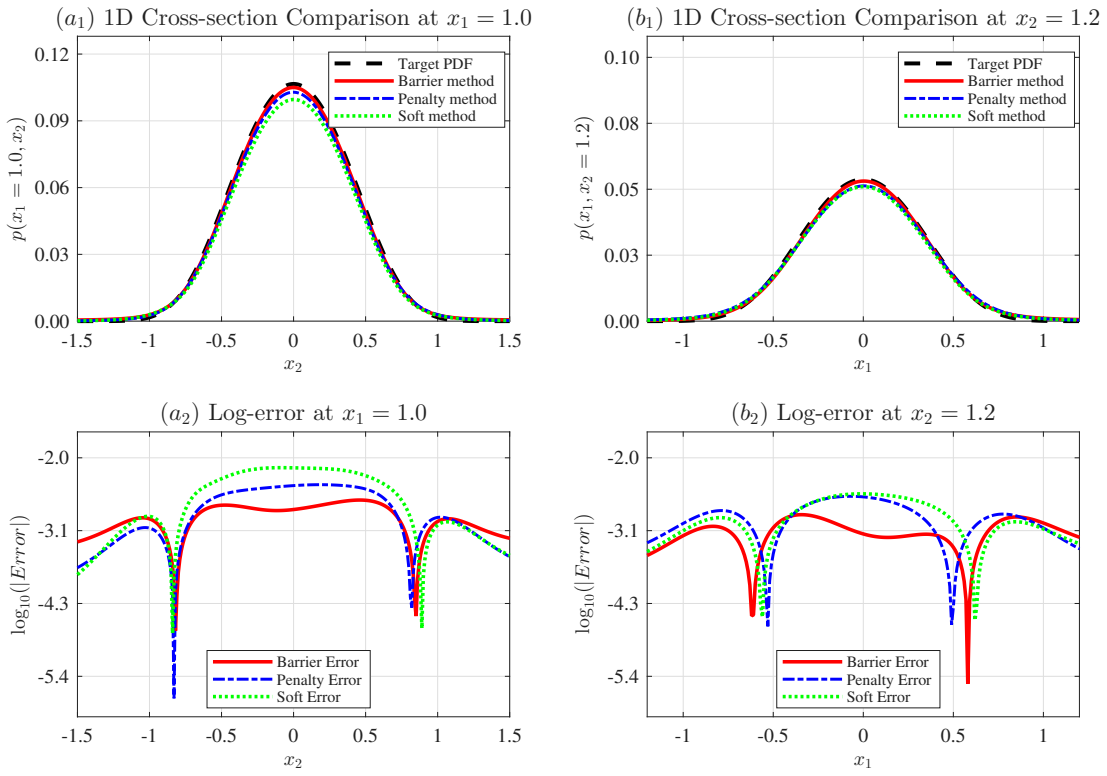


Fig. 4: Cross-sectional profiles and pointwise absolute PDF errors along selected cross-sections for the first SDOF target PDF. Sharp minima in logarithmic error curves may occur at intersections of the predicted and target profiles.

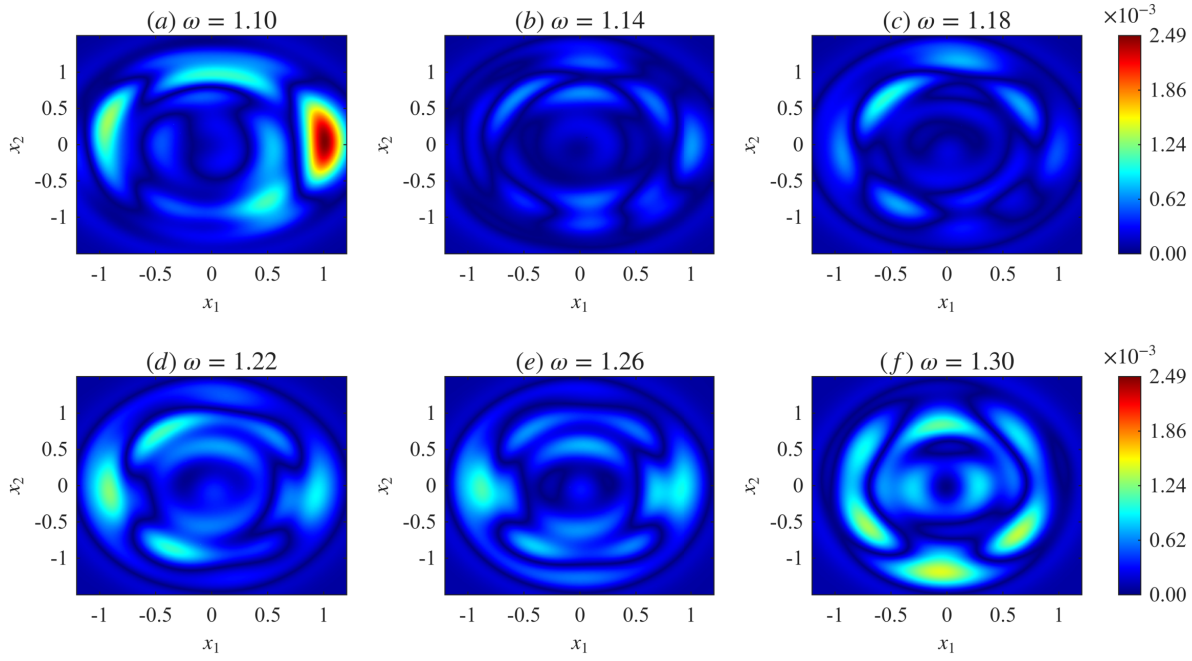


Fig. 5: Pointwise absolute PDF errors from the same trained barrier density network for the first SDOF target at $\omega = 1.10, 1.14, 1.18, 1.22, 1.26, 1.30$, without parameter-specific retraining. These evaluation values lie within the training parameter range $[1.1, 1.3]$, including its endpoints; training residual parameters are sampled uniformly rather than restricted to this evaluation grid.

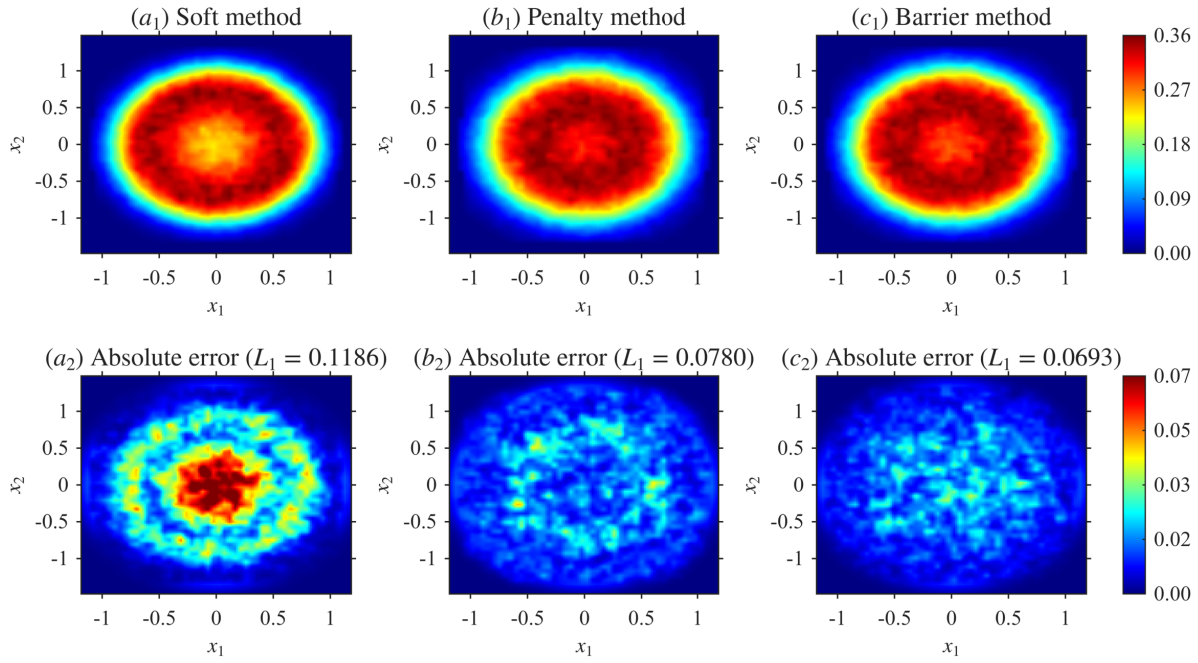


Fig. 6: MC validation of the learned feedback controls for the first SDOF target PDF at $\omega = 1.18$. The upper row shows the MC-estimated stationary PDFs for the soft constraint, penalty, and barrier formulations, and the lower row shows the corresponding pointwise absolute errors relative to the prescribed target PDF, with the L_1 errors reported in each panel.

Table 2 reports the wall-clock training time for the three methods in the first SDOF benchmark. The barrier method requires the largest computational cost, the penalty method has an intermediate cost, and the soft constraint method is the least expensive.

4.1.2. The second target PDF

The specified target stationary probability density function is

$$p_{\text{target}}(x_1, x_2) = C \exp\left[x_2^2 - 4x_2^4 - (2x_1^2 + \frac{8}{3}x_1^3 + 2x_1^4)\right]. \tag{26}$$

The computations for this fixed target PDF are performed on $\Omega_{\mathbf{x}} = [-1.2, 1.2] \times [-1.5, 1.5]$. This target does not involve a varying target-distribution parameter.

Fig. 7 examines the second SDOF target, whose cubic terms create an asymmetric non-Gaussian density. All three methods locate the dominant concentration region, while the pointwise absolute PDF error maps show differences in agreement with the target over the plotted domain.

Fig. 8 presents representative one-dimensional cross-sections for the second SDOF target together with the corresponding pointwise absolute errors. The displayed profiles allow comparison of the asymmetric peak and the decay on both sides of the target density. Any smaller pointwise absolute errors are specific to the displayed coordinates and do not imply improved relative accuracy in low-probability regions.

Fig. 9 complements the network-density reconstruction by showing the dominant asymmetric concentration in the simulated responses. The reported L_1 errors are 0.195, 0.0942, and 0.0891 for the soft, penalty, and barrier formulations, respectively.

4.2. Stochastic levitation system

This example introduces a nonlinear physical input: the total current enters the electromagnetic force quadratically and requires selection of a physically relevant current branch. The two targets are assessed on their stated finite

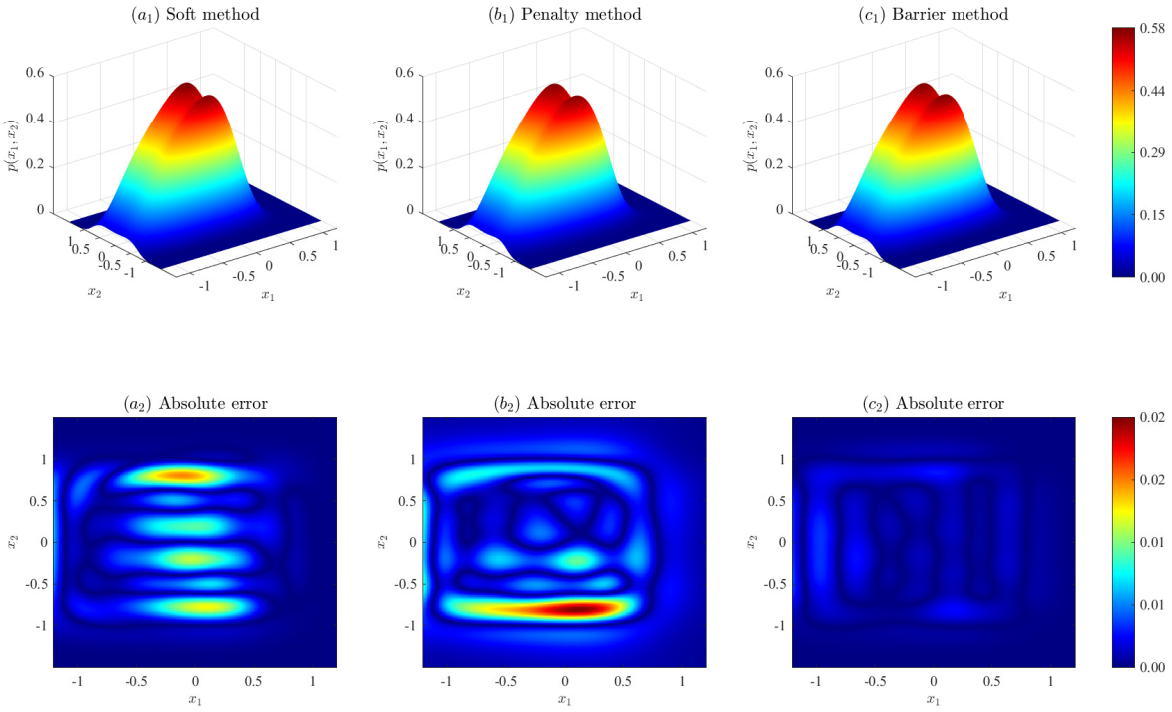


Fig. 7: Comparison of learned stationary PDFs and pointwise absolute errors for the second SDOF target PDF.

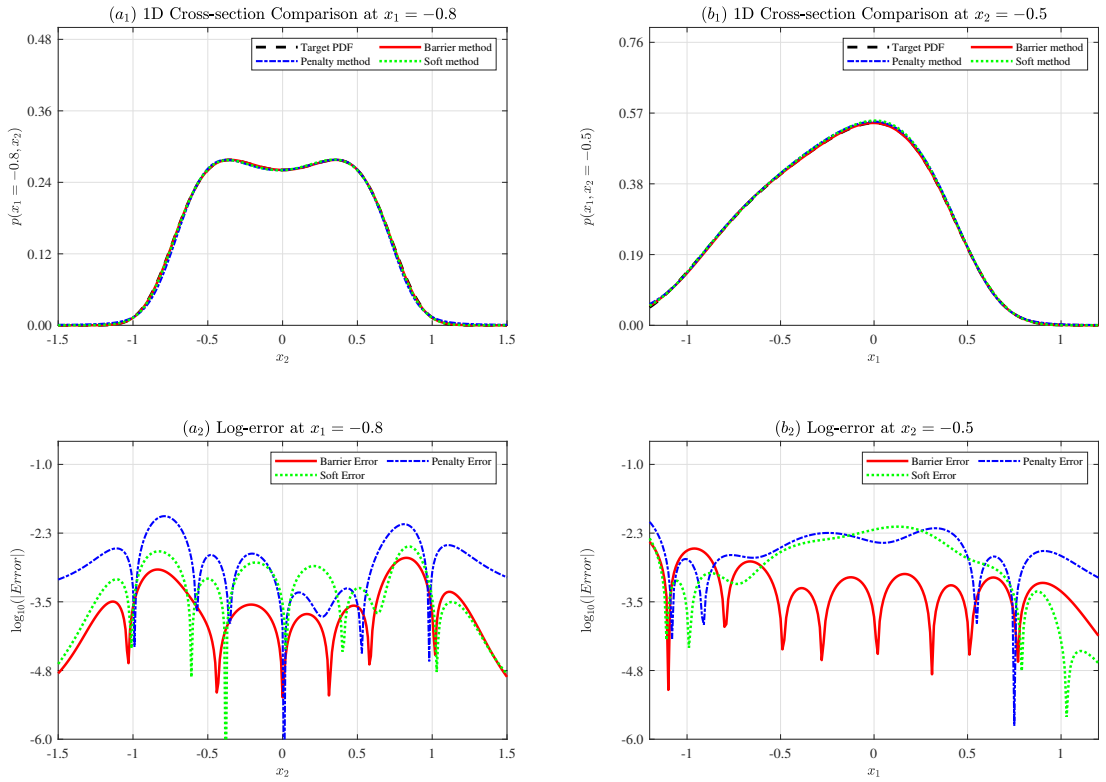


Fig. 8: Cross-sectional profiles and pointwise absolute PDF errors along selected cross-sections for the second SDOF target PDF.

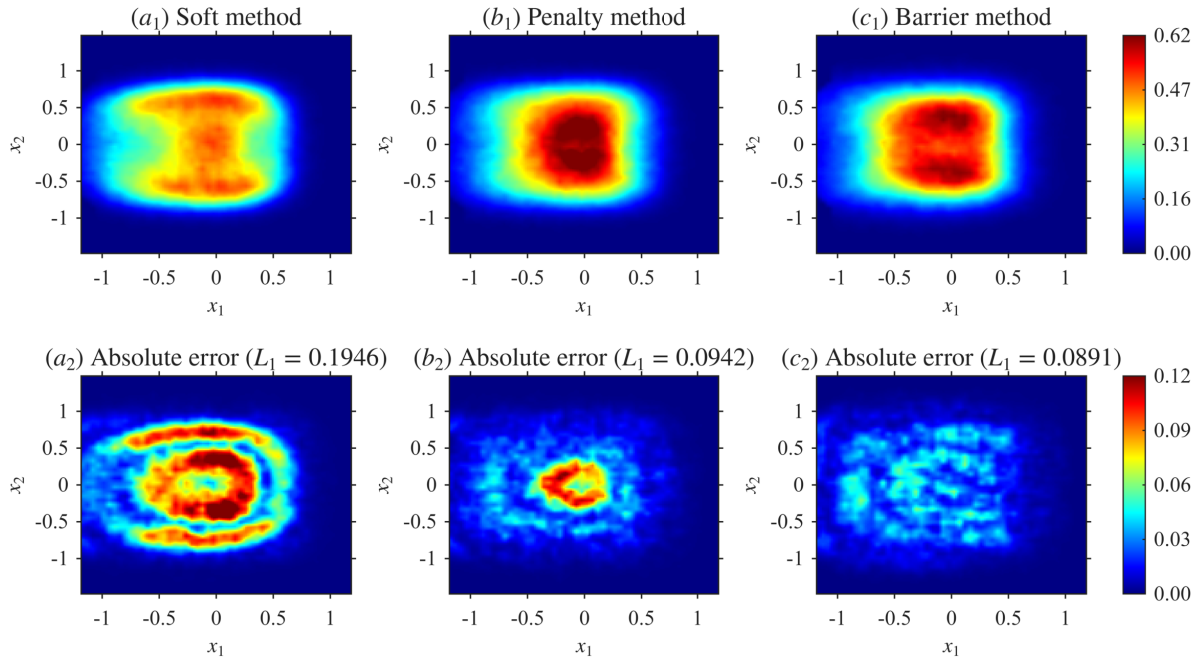


Fig. 9: MC validation of the learned feedback controls for the second SDOF target PDF. The upper row shows the MC-estimated stationary PDFs for the soft constraint, penalty, and barrier formulations, and the lower row shows the corresponding pointwise absolute errors relative to the prescribed asymmetric target PDF, with the L_1 errors reported in each panel.

computational domains. Consider the stochastic levitation system. The single-point electromagnetic levitation system is modeled as

$$\begin{aligned}\dot{X}_1 &= X_2,\\ \dot{X}_2 &= \frac{g}{z_0^2} - \frac{\mu_0 N^2 S}{4mz_0^2}\left[\frac{k_p z_0^2 X_1 + k_v z_0^2 X_2 + u(X_1,X_2) + i_0}{z_0^2 X_1 + z_0}\right]^2 + \frac{\zeta(t)}{mz_0^2},\end{aligned} \tag{27}$$

where $\zeta(t)$ is a zero-mean Gaussian white-noise process satisfying $\mathbb{E}[\zeta(t)\zeta(s)] = D_G\delta(t-s)$, with D_G denoting the noise intensity.

The corresponding stationary FPK equation is

$$0 = -\frac{\partial(x_2 p)}{\partial x_1} - \frac{\partial}{\partial x_2}\left\{\left[\frac{g}{z_0^2} - \frac{\mu_0 N^2 S}{4mz_0^2}\left(\frac{k_p z_0^2 x_1 + k_v z_0^2 x_2 + u(x_1,x_2) + i_0}{z_0^2 x_1 + z_0}\right)^2\right] p\right\} + \frac{D_G}{2m^2 z_0^4}\frac{\partial^2 p}{\partial x_2^2}. \tag{28}$$

Due to the significant differences in the orders of magnitude of the system parameters, directly solving Eq.(28) is numerically challenging [19, 46]. To reduce this scale disparity, Eq.(28) is first multiplied by z_0^2, and the derivative with respect to x_2 is then expanded, yielding

$$\begin{aligned}0 = {} & 2\frac{\mu_0 N^2 S}{4mz_0^2}\frac{\left[k_v z_0^2 + \frac{\partial}{\partial x_2}u(x_1,x_2)\right]}{(z_0 x_1 + 1)^2}\left[k_p z_0^2 x_1 + k_v z_0^2 x_2 + u(x_1,x_2) + i_0\right] p\\ & - z_0^2 x_2\frac{\partial p}{\partial x_1} - \left[g - \frac{\mu_0 N^2 S}{4mz_0^2}\left(\frac{k_p z_0^2 x_1 + k_v z_0^2 x_2 + u(x_1,x_2) + i_0}{z_0 x_1 + 1}\right)^2\right]\frac{\partial p}{\partial x_2} + \frac{D_G}{2m^2 z_0^2}\frac{\partial^2 p}{\partial x_2^2}.\end{aligned} \tag{29}$$

4.2.1. The first target PDF

The first scenario considers control over the vibration amplitude, which is characterized here through the variance of the levitation gap [19, 20]. The target PDF can be designed as

$$p_1(x_1, x_2) = C_1(\omega) \exp[-\phi(x_1, x_2)] = C_1(\omega) \exp\left\{-\frac{1}{2}\omega\left(Q_1^2 x_1^2 + x_2^2\right)\right\}, \tag{30}$$

where $C_1(\omega)$ is the parameter-dependent normalization constant, ω is an adjustable target-distribution parameter, and $Q_1^2 = 500$. The parameter ω controls the overall concentration of the target density, while Q_1 determines the relative scaling between the two state directions. The computations for this target are performed on $\Omega_{\mathbf{x}} = [-0.2, 0.2] \times [-4, 4]$, with $\omega \in [4.8, 5.3]$.

Table 3: Physical parameters of the stochastic levitation model.

N	m	μ_0	S	i_0	z_0	k_p	k_v	g	D_G
450	700	$4\pi \times 10^{-7}$	0.024	21.2075	0.01	4823.043	76.4208	9.81	1.0

Fig. 10 illustrates the first levitation case, where the target density is designed to regulate the response variance of the levitation gap. The predicted PDFs reproduce the narrow concentration band imposed by the target distribution, while the pointwise absolute PDF error maps show differences in density agreement over the plotted domain.

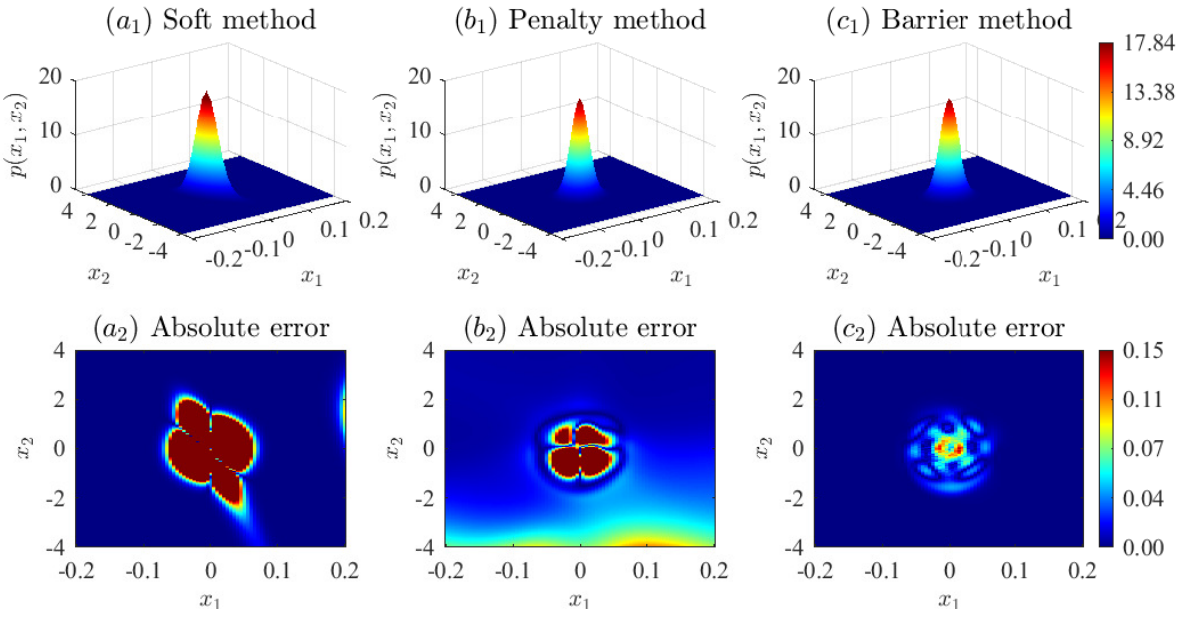


Fig. 10: Comparison of learned stationary PDFs and pointwise absolute errors for the first levitation target PDF.

Fig. 11 compares representative PDF cross-sectional profiles of the narrow target density and the corresponding pointwise absolute PDF errors. These sections assess agreement with the target density curve at the displayed coordinates, rather than the variance of the response. The Gaussian target density is strictly positive at finite coordinates, although its values away from the peak can be extremely small. Small absolute errors there do not establish accurate relative errors or tail probabilities. Sharp minima in logarithmic error curves can result from intersections between predicted and target curves.

Fig. 12 evaluates one trained barrier density network at the six parameter values listed in the caption. These plots assess density approximation at displayed values, not interval-wide closed-loop control accuracy or the parameter-scan performance of all three strategies.

Fig. 13 reports closed-loop density agreement for the first levitation target on the common normalized comparison domain. The L_1 errors are 0.43, 0.282, and 0.11 for the soft, penalty, and barrier formulations, respectively.

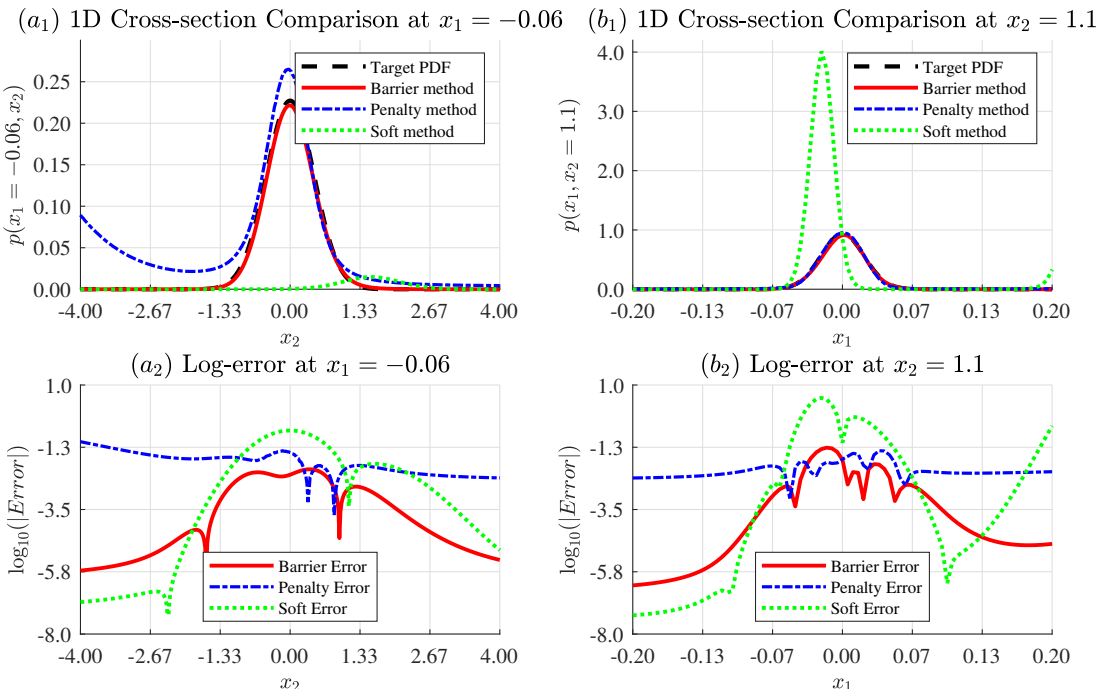


Fig. 11: Cross-sectional profiles and pointwise absolute PDF errors for the first levitation target PDF. Sharp minima in logarithmic error curves may occur at intersections of the predicted and target profiles.

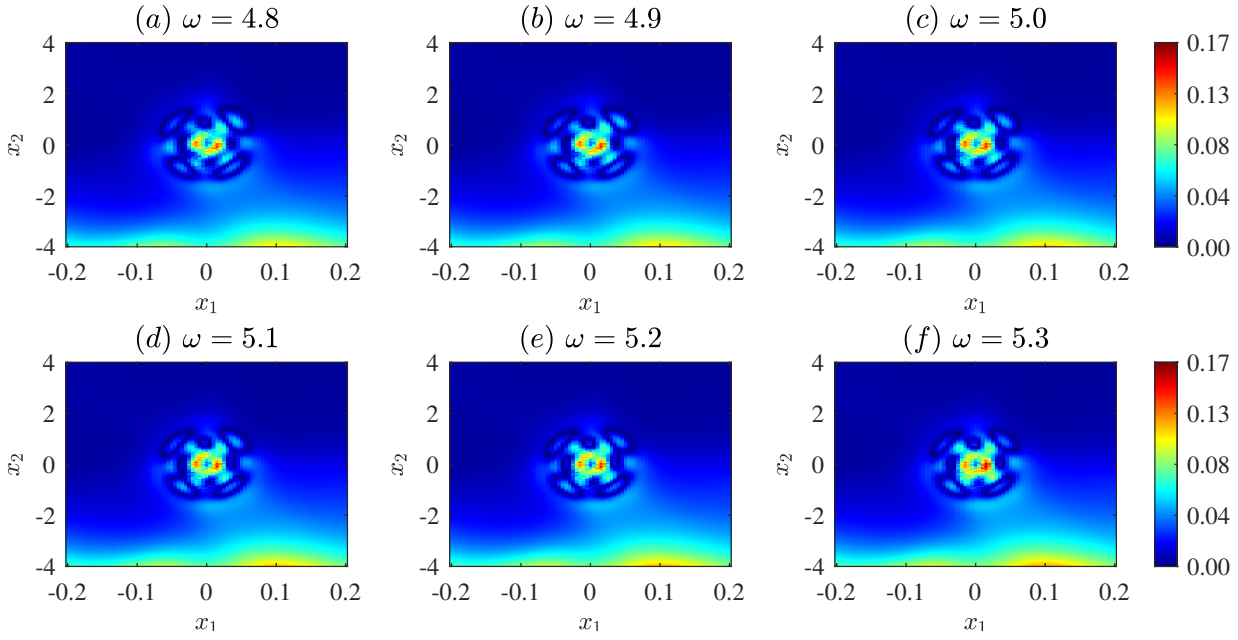


Fig. 12: Parameter sweep of the stationary-density approximation obtained using the barrier method for the first target distribution of the electromagnetic levitation system. The same trained density network is evaluated at $\omega = 4.8, 4.9, \ldots, 5.3$ without retraining.

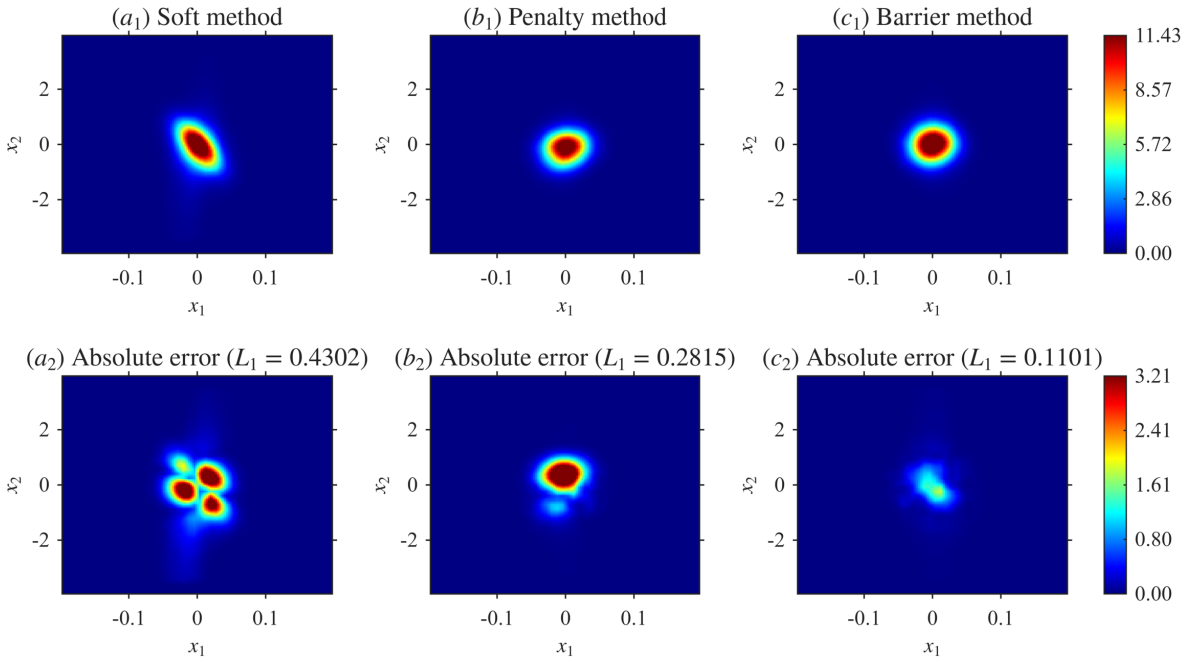


Fig. 13: MC validation of the learned feedback controls for the first levitation target PDF at $\omega = 5.0$. The upper row shows the MC-estimated stationary PDFs for the soft constraint, penalty, and barrier formulations, and the lower row shows the corresponding pointwise absolute errors relative to the prescribed target PDF, with the L_1 errors reported in each panel.

4.2.2. The second target PDF

Based on the first scenario, this section further considers controlling the mean levitation gap. Specifically, the desired mean position of the levitation gap is shifted, and the target PDF is assigned a nonzero mean. The second target PDF is defined as

$$p_2(x_1, x_2) = C_2(\omega) \exp[-\phi(x_1, x_2)] = C_2(\omega) \exp\left\{-\frac{1}{2}\omega \left[Q_2^2(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2\right]\right\}, \tag{31}$$

where $C_2(\omega)$ is the parameter-dependent normalization constant, and μ_1 and μ_2 determine the prescribed mean values in the x_1 and x_2 directions, respectively. The parameter ω is varied over the prescribed range and controls the overall concentration of the target density, whereas Q_2 is fixed and determines the relative scaling between the two state directions. The computations for this target are performed on $\Omega_{\mathbf{x}} = [1.98, 2.02] \times [-1, 1]$, with $\omega \in [9.8, 10.3]$. In this example, $Q_2^2 = 8000$, $\mu_1 = 2.0$, and $\mu_2 = 0.0$.

Fig. 14 focuses on the second levitation target, in which the desired mean levitation gap is shifted away from the origin. The three methods place the probability mass around the prescribed offset, but the pointwise PDF error maps reveal different levels of accuracy around the shifted peak. The penalty strategy improves over the soft constraint baseline, while the barrier method most effectively localizes the remaining error near the peak region.

Fig. 15 compares cross-sectional profiles and pointwise absolute PDF errors for the shifted levitation target. On the displayed sections, the barrier method more closely tracks the peak height and location.

Fig. 16 shows simulated response concentration near $x_1 = 2.0$. The reported L_1 errors are 0.0952, 0.097, and 0.086 for the soft, penalty, and barrier formulations, respectively.

4.3. Two-degree-of-freedom (2-DOF) Duffing system

This example extends the approximation to a four-dimensional state space. The network-density plots are fixed-coordinate slices, whereas the MC plots are two-dimensional marginals obtained from the simulated responses. Slices and marginals test different aspects of the density and do not by themselves verify the full four-dimensional joint distribution. Consider the coupled Duffing oscillator excited by Gaussian white noise. Using the state vector $\mathbf{X} =$

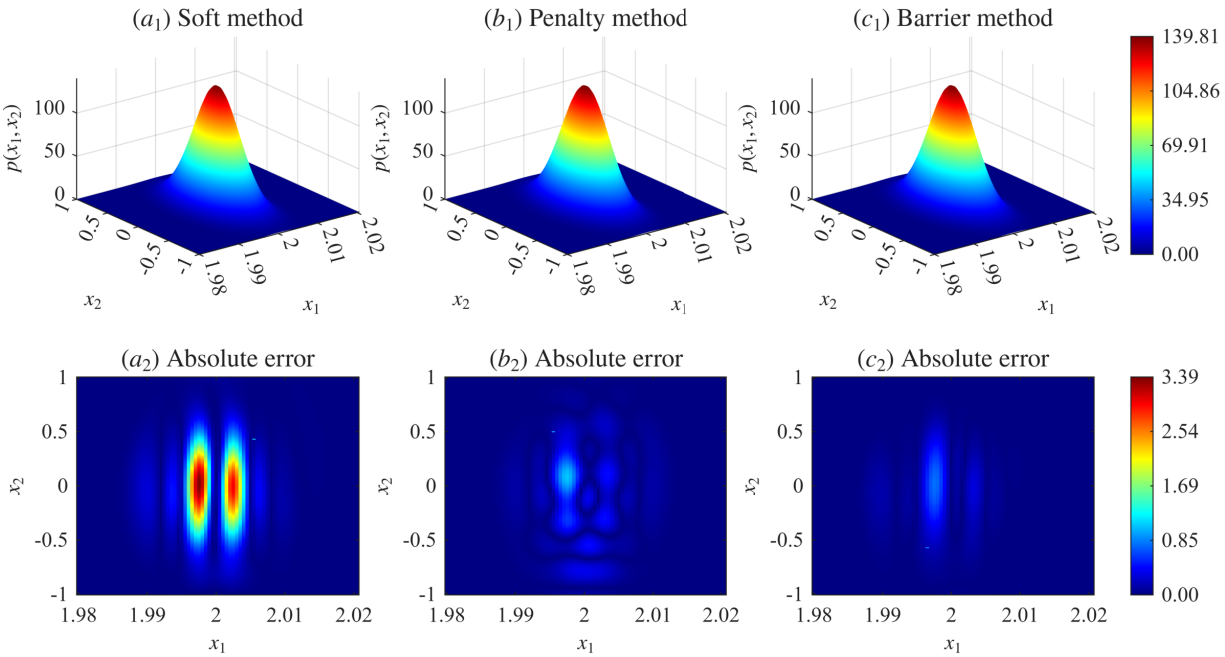


Fig. 14: Comparison of learned stationary PDFs and pointwise absolute errors for the second levitation target PDF.

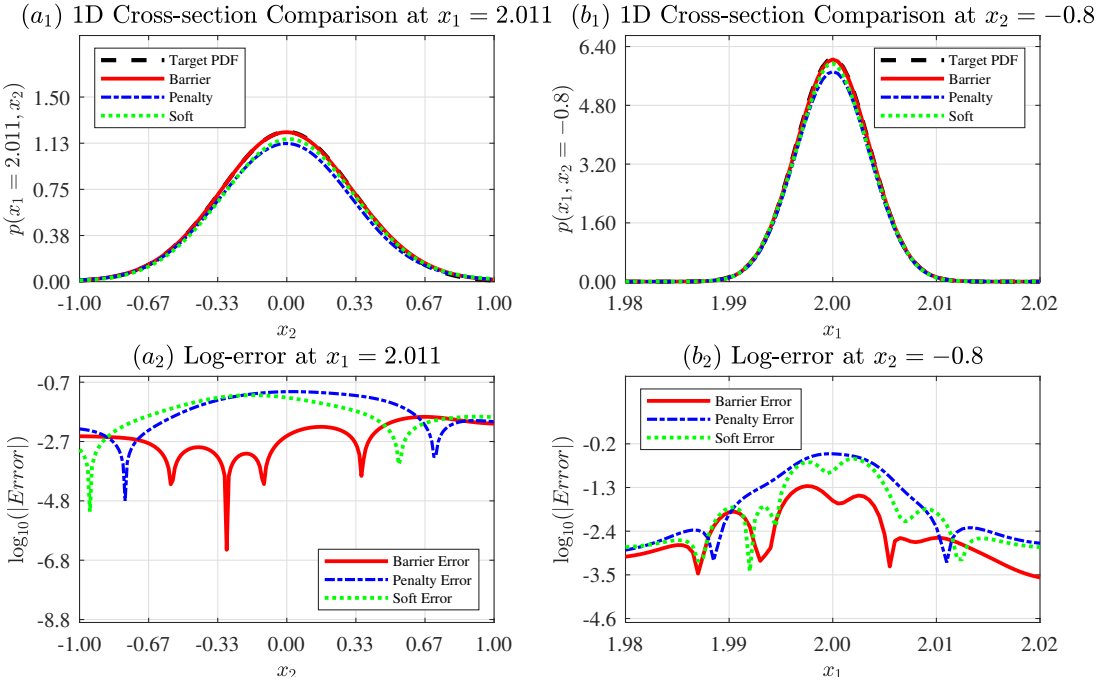


Fig. 15: Cross-sectional profiles and pointwise absolute PDF errors for the second levitation target PDF.

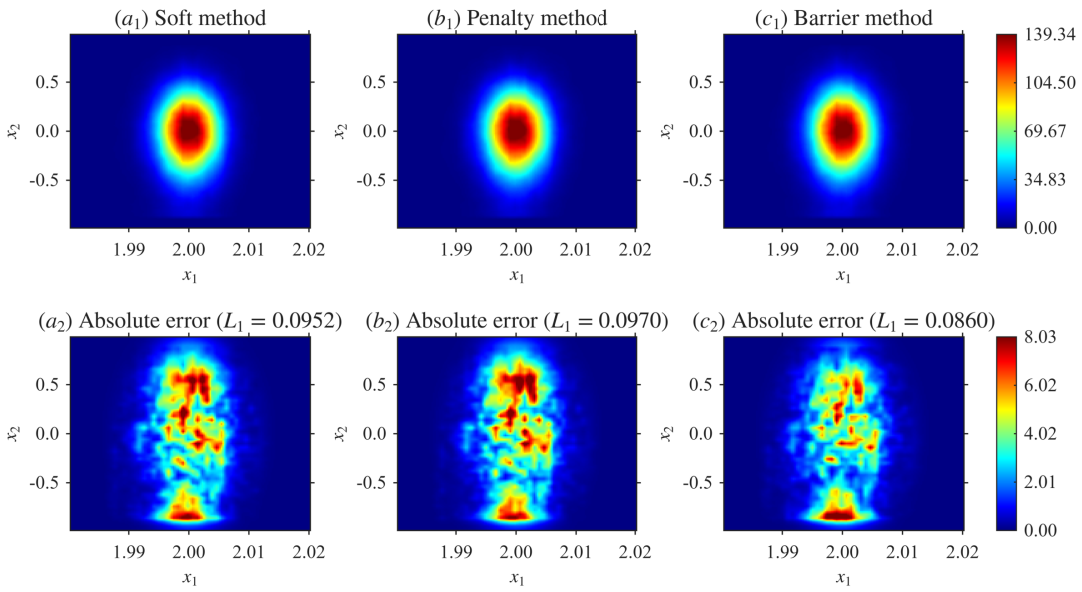


Fig. 16: MC validation of the learned feedback controls for the second levitation target PDF at $\omega = 10.3$. The upper row shows the MC-estimated stationary PDFs for the soft constraint, penalty, and barrier formulations, and the lower row shows the corresponding pointwise absolute errors relative to the prescribed shifted target PDF, with the L_1 errors reported in each panel.

$(X_1, X_2, X_3, X_4)^{\mathrm{T}}$, where X_1 and X_3 are displacement-type coordinates and X_2 and X_4 are the corresponding velocity-type coordinates, the equation of motion is written as

$$\begin{aligned} \mathrm{d}X_1 &= X_2\,\mathrm{d}t, \\ \mathrm{d}X_2 &= \left(-\beta_1 X_2 - \frac{\partial H}{\partial X_1} - u_1\right)\mathrm{d}t + \sqrt{2D_1}\,\mathrm{d}w_1(t), \\ \mathrm{d}X_3 &= X_4\,\mathrm{d}t, \\ \mathrm{d}X_4 &= \left(-\beta_2 X_4 - \frac{\partial H}{\partial X_3} - u_2\right)\mathrm{d}t + \sqrt{2D_2}\,\mathrm{d}w_2(t), \end{aligned} \tag{32}$$

where the potential function is

$$H(\mathbf{X}) = \frac{A_1}{2}X_1^2 + \frac{A_2}{2}X_3^2 + \frac{\alpha_1}{4}X_1^4 + \frac{\alpha_2}{4}X_3^4 + \alpha_3 X_1^2 X_3^2. \tag{33}$$

The parameters are $A_1 = -0.1$, $A_2 = 0.1$, $\alpha_1 = \alpha_2 = 10$, $\alpha_3 = 0.001$, $\beta_1 = \beta_2 = 0.09$, and $D_1 = D_2 = 0.002$. Here, β_i are the linear damping coefficients, $w_i(t)$ are independent standard Wiener processes, and $2D_i$ denote the noise intensities.

4.3.1. The first target PDF

The first target PDF is set as

$$p_{\mathrm{target}}(x_1, x_2, x_3, x_4) = C_0(\omega)\exp\left[-\frac{1}{2}\left(x_1^2 + x_2^2\right) - \frac{1}{2}\left(x_3^2 + \omega x_4^2\right)\right], \tag{34}$$

where $C_0(\omega)$ is the parameter-dependent normalization constant and $\omega > 0$. The parameter ω controls the spread of the target density along x_4. The computations are performed on $\Omega_{\mathbf{x}} = [-3, 3]^4$, with $\omega \in [1.8, 2.3]$. In the stationary-density comparisons below, $x_3 = x_4 = 0$ and the four-dimensional density is evaluated on the (x_1, x_2) plane. The target slice is $p_{\mathrm{target}}(x_1, x_2, 0, 0) = C_0(\omega)\exp[-(x_1^2 + x_2^2)/2]$. Thus, ω affects its amplitude through $C_0(\omega)$, while its Gaussian width along x_1 and x_2 remains unchanged.

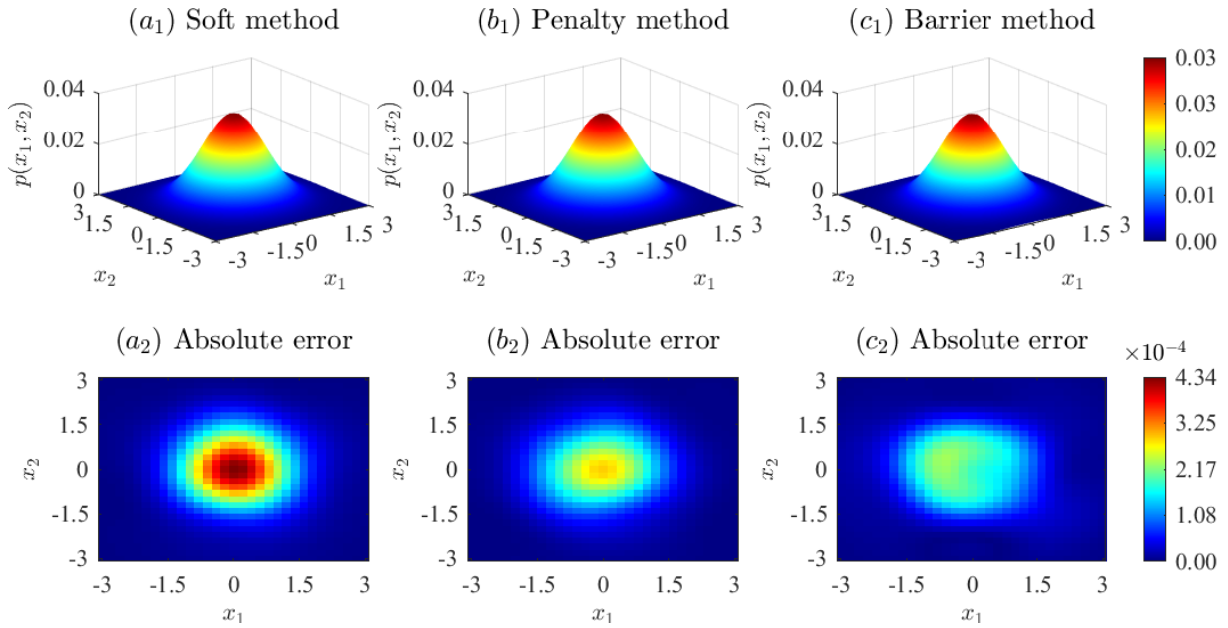


Fig. 17: Learned four-dimensional stationary PDFs evaluated on the (x_1, x_2) slice with $x_3 = x_4 = 0$ for the first Duffing target. The upper row shows the soft constraint, penalty, and barrier predictions; the lower row shows their pointwise absolute errors relative to the target slice.

Fig. 17 assesses the centered Gaussian density on the slice $x_3 = x_4 = 0$. The displayed absolute errors are concentrated near the central peak; this comparison concerns a slice of the four-dimensional density.

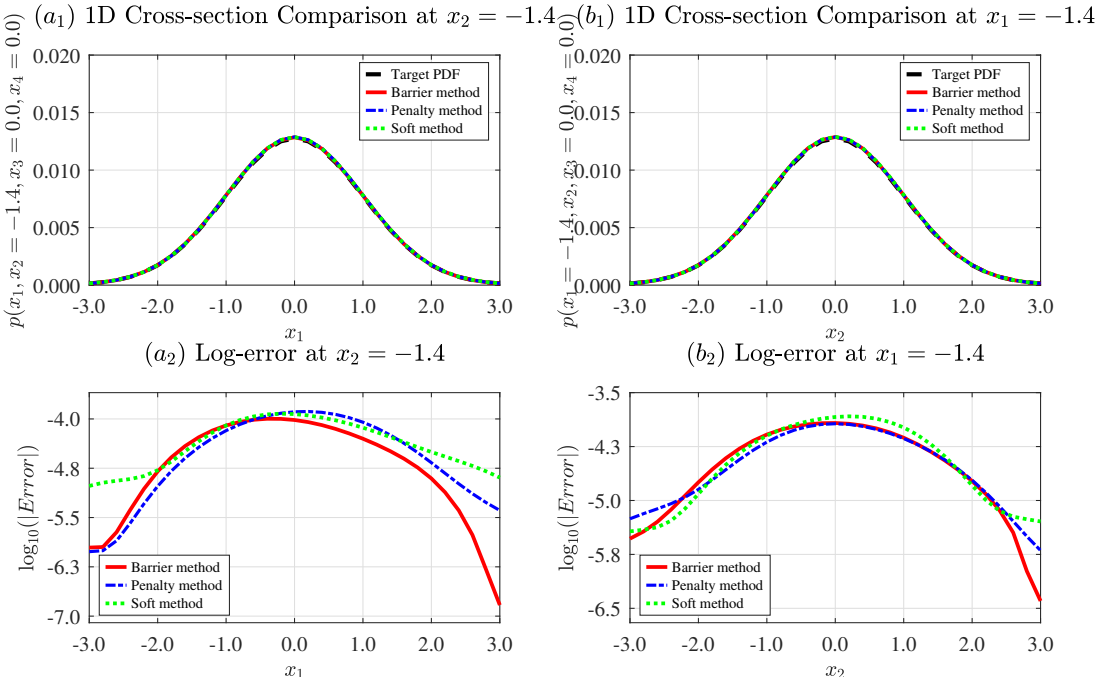


Fig. 18: One-dimensional profiles of the (x_1, x_2) density slice with $x_3 = x_4 = 0$ for the first Duffing target. The left column varies x_1 at $x_2 = -1.4$, and the right column varies x_2 at $x_1 = -1.4$. The upper panels compare the target and learned density values; the lower panels show $\log_{10}$ of the pointwise absolute errors.

Fig. 18 examines two off-center cuts of the same slice: $p(x_1, -1.4, 0, 0)$ and $p(-1.4, x_2, 0, 0)$. The learned profiles closely follow the target curves in both directions, including the peak near zero in the varying coordinate and the decay toward the tails. The logarithmic error curves reveal differences that are difficult to distinguish in the upper panels.

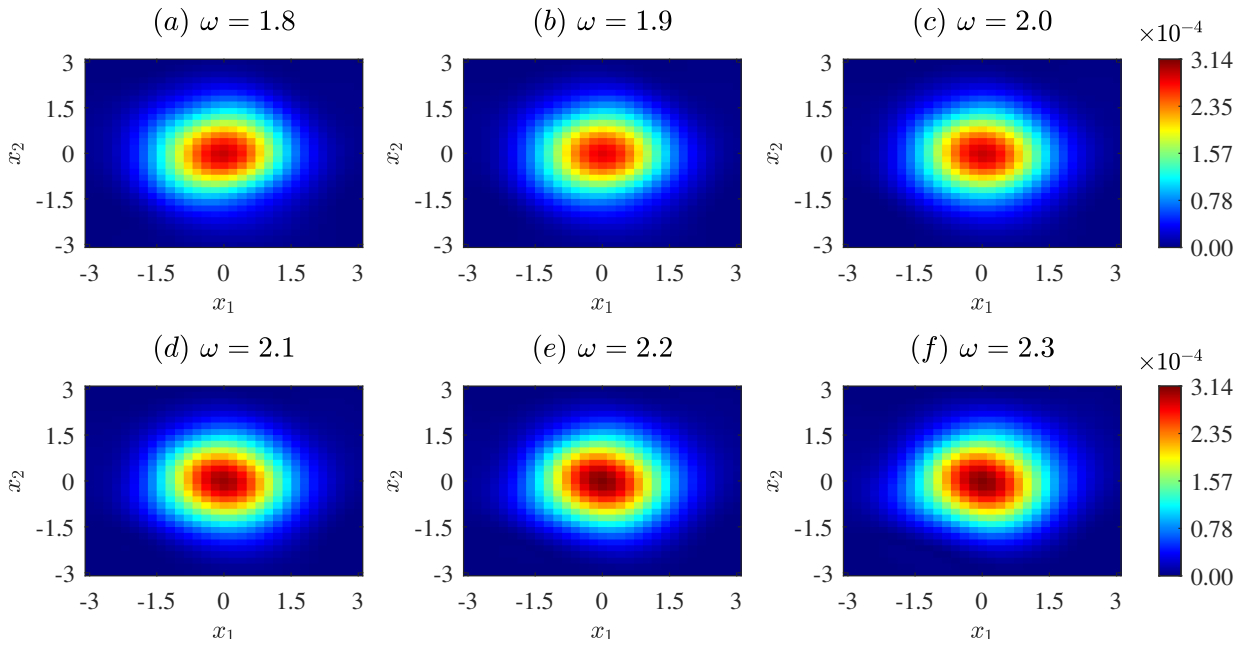


Fig. 19: Pointwise absolute PDF errors from the same trained barrier density network for the first Duffing target on the slice $x_3 = x_4 = 0$ at $\omega = 1.80, 1.90, 2.00, 2.10, 2.20, 2.30$. The evaluation values lie within the training parameter range $[1.8, 2.3]$, including its endpoints.

Fig. 19 evaluates one trained barrier density network at six displayed values in the training parameter range. Errors on the slice $x_3 = x_4 = 0$ are concentrated near the origin and remain of order 10^{-4} on the common scale.

Fig. 20 presents the MC validation at $\omega = 1.8$ using two-dimensional marginal PDFs of the controlled responses. All three methods reproduce the centered concentration in both coordinate planes. The (x_1, x_2) marginal gives L_1 errors of 0.1055, 0.1030, and 0.1026 for the soft constraint, penalty, and barrier formulations, respectively. The corresponding (x_3, x_4) errors are 0.0756, 0.0806, and 0.0830. The barrier formulation has the lowest reported error in the first marginal, whereas the soft constraint formulation has the lowest error in the second. These marginal comparisons provide closed-loop statistical validation at the selected parameter value, without establishing a uniform ranking of the methods or verifying the complete four-dimensional joint density.

4.3.2. The second target PDF

$$p_{\text{target}}(x_1, x_2, x_3, x_4; \omega) = C_0(\omega) \exp\left(-\frac{x_1^4}{4} - \frac{x_2^2}{2} - \frac{\omega x_3^2}{2} - \frac{x_4^2}{2}\right),$$
$$C_0(\omega) = \frac{\sqrt{2\omega}}{\Gamma(1/4)(2\pi)^{3/2}}, \tag{35}$$

where Γ denotes the gamma function, $C_0(\omega)$ normalizes the target density over $\mathbb{R}^4$, and $\omega > 0$. The quartic term $-x_1^4/4$ produces a non-Gaussian distribution in the x_1 direction, whereas x_2, x_3, and x_4 have Gaussian target distributions. The parameter ω controls the variance in the x_3 direction, $\text{Var}(X_3) = 1/\omega$, while the full-space target variances in the x_2 and x_4 directions are both one. In the following stationary-equation comparisons, x_1 and x_2 are fixed at zero and the learned four-dimensional PDF is evaluated on the (x_3, x_4) plane. The resulting target slice is $p_{\text{target}}(0, 0, x_3, x_4; \omega) = C_0(\omega) \exp[-(\omega x_3^2 + x_4^2)/2]$; the quartic dependence on x_1 is therefore not visible on this slice. The computations for this target are performed on $\Omega_{\mathbf{x}} = [-3, 3]^4$, with $\omega \in [0.8, 1.3]$. For the finite-domain MC error comparisons, the target and MC-estimated densities follow the common normalization convention in Eq.(20).

An analytical feedback pair can be constructed for this target. With the negative control signs in Eq.(32), one such

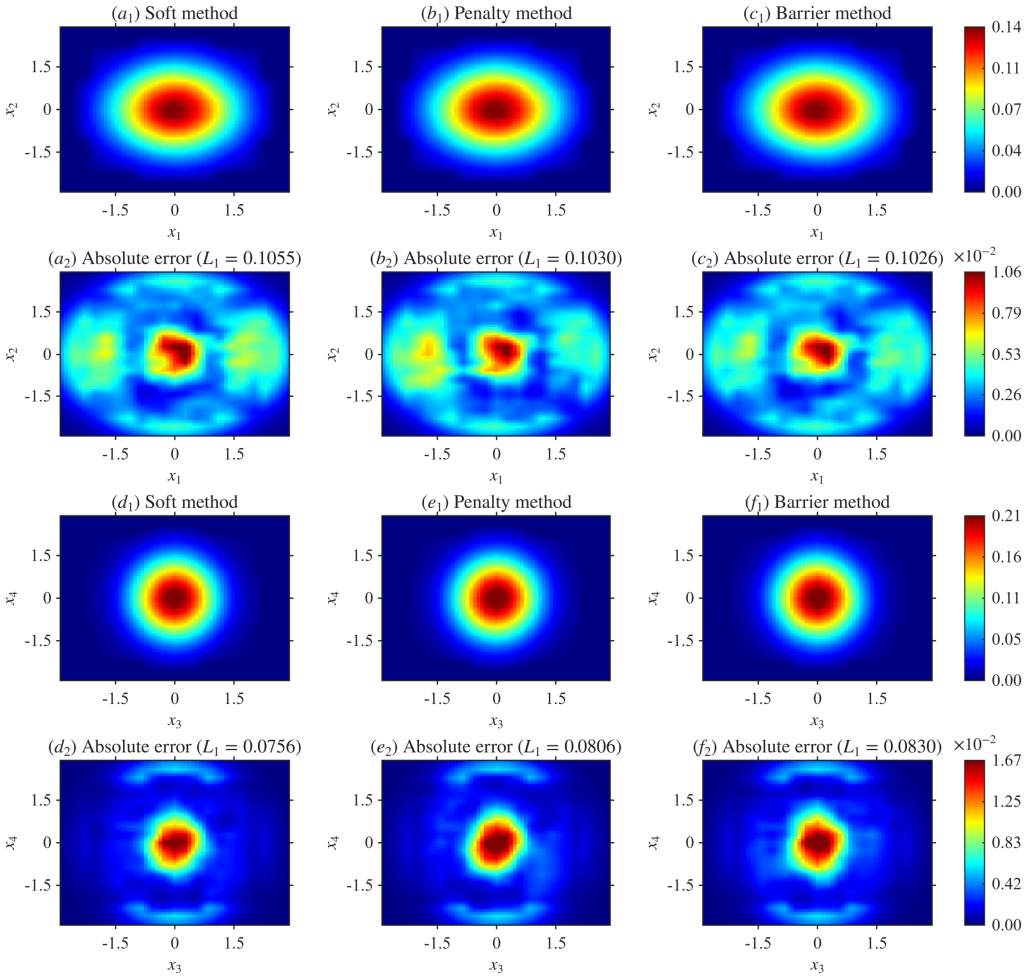


Fig. 20: MC validation of the learned feedback controls for the first Duffing target PDF at $\omega = 1.8$. The first two rows show the MC-estimated marginal PDFs in the (x_1, x_2) plane and their corresponding pointwise absolute errors, while the last two rows show the results in the (x_3, x_4) plane. The L_1 errors are reported in the corresponding error panels.

pair is

$$u_1^{\mathrm{ref}}(\mathbf{x};\omega) = x_1^3 + (D_1 - \beta_1)x_2 - \frac{\partial H}{\partial x_1},$$
$$u_2^{\mathrm{ref}}(\mathbf{x};\omega) = \omega x_3 + (D_2 - \beta_2)x_4 - \frac{\partial H}{\partial x_3}.$$

The resulting drift components in the X_2 and X_4 equations are $-x_1^3 - D_1 x_2$ and $-\omega x_3 - D_2 x_4$, respectively. Direct substitution into the stationary FPK equation shows that the Hamiltonian transport terms cancel and that the damping and diffusion terms balance for the density in Eq.(35). This provides an analytical construction compatible with the target; it does not establish that the feedback pair is unique. The neural networks are trained without this reference law and need not recover the same pair to approximate the prescribed stationary density. No pointwise control comparison with this pair is used in the reported numerical validation.

Fig. 21 evaluates the reconstruction accuracy of the second Duffing target at $\omega = 1.0$ on the (x_3, x_4) slice with $x_1 = x_2 = 0$. The target slice is $C_0(1.0)\exp[-(x_3^2+x_4^2)/2]$, which is Gaussian in both displayed directions and has equal widths at this parameter value. The non-Gaussian dependence on x_1 is not assessed by this slice. All three methods reproduce the central concentration and the principal shape of this slice. The pointwise errors are concentrated mainly in the central region and are of order 10^{-4} on the displayed common scale. Their spatial patterns differ, with an asymmetric error pattern for the penalty formulation and more centrally concentrated patterns for the other two formulations. This figure compares the methods at a fixed parameter value.

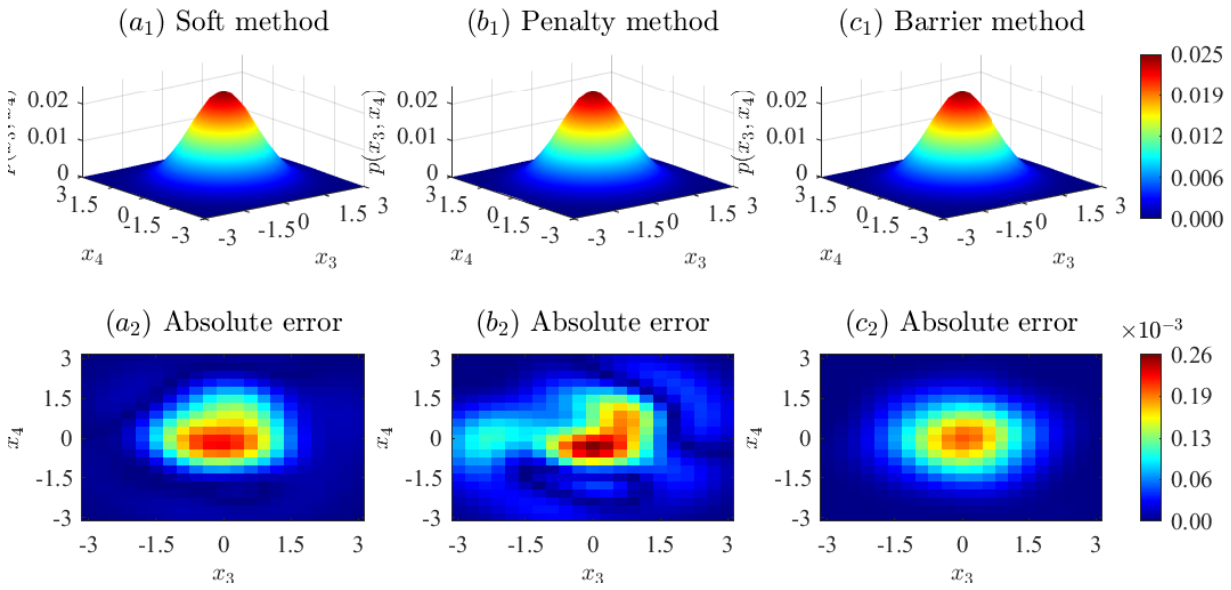


Fig. 21: Selected two-dimensional slice of the learned four-dimensional stationary PDF on the (x_3, x_4) plane with $x_1 = x_2 = 0$ for the second Duffing target at $\omega = 1.0$. The upper row shows the learned density slices, and the lower row shows their pointwise absolute errors relative to the Gaussian target slice $C_0(1.0)\exp[-(x_3^2 + x_4^2)/2]$.

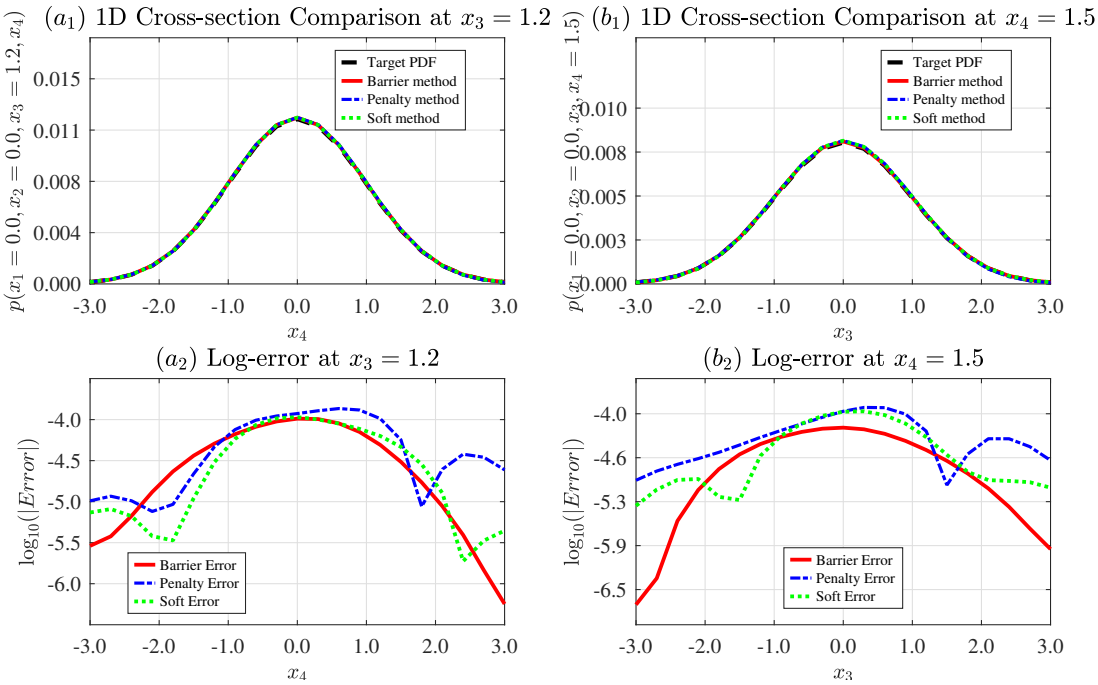


Fig. 22: Cross-sectional profiles of the (x_3, x_4) slice with $x_1 = x_2 = 0$ for the second Duffing target at $\omega = 1.0$. The left column varies x_4 at $x_3 = 1.2$, and the right column varies x_3 at $x_4 = 1.5$. Both target profiles are Gaussian. The upper panels compare the target and learned density values, and the lower panels show $\log_{10}$ of the pointwise absolute errors.

Fig. 22 presents two off-center one-dimensional cuts at $\omega = 1.0$, with $x_1 = x_2 = 0$. The left column shows $p(0, 0, 1.2, x_4)$, which has a Gaussian profile along x_4; the right column shows $p(0, 0, x_3, 1.5)$, which also has a Gaussian profile. The corresponding target curves are $C_0(1.0)\exp[-(1.2^2 + x_4^2)/2]$ and $C_0(1.0)\exp[-(x_3^2 + 1.5^2)/2]$, respectively. The learned profiles closely follow the target curves on both cuts. The lower panels display the pointwise absolute errors on a base-10 logarithmic scale. The error curves cross locally, so their ordering depends on the displayed coordinate

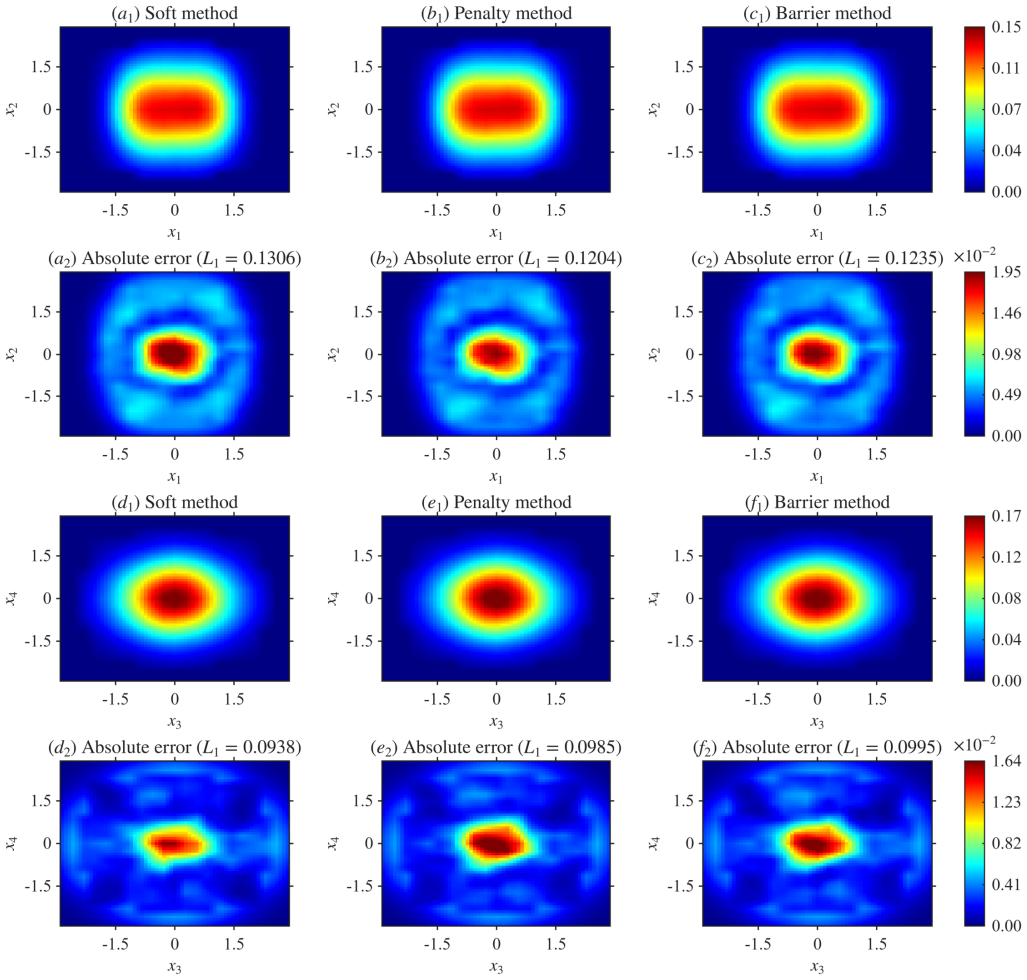


Fig. 23: MC validation of the learned feedback controls for the second Duffing target PDF at $\omega = 1.0$. The first two rows show the MC-estimated marginal PDFs in the (x_1, x_2) plane and their corresponding pointwise absolute errors, while the last two rows show the results in the (x_3, x_4) plane. The L_1 errors are reported in the corresponding error panels.

Fig. 23 shows the MC-estimated marginal PDFs for the second Duffing target. The target (x_1, x_2) marginal retains the quartic dependence on x_1 and the Gaussian dependence on x_2, whereas the target (x_3, x_4) marginal is Gaussian in both coordinates. At $\omega = 1.0$, the target widths along x_3 and x_4 are equal. In the (x_1, x_2) marginal, the three methods reproduce the broad central concentration, with L_1 errors of 0.131, 0.120, and 0.124 for the soft constraint, penalty, and barrier formulations, respectively; the penalty formulation gives the smallest error in this marginal. In the (x_3, x_4) marginal, the controlled responses reproduce the centered Gaussian target concentration, with L_1 errors of 0.0938, 0.0985, and 0.0995; the soft constraint formulation gives the smallest error in this marginal. The change in the relative ordering across the two marginals again indicates that the MC results should be used primarily to verify the controlled stationary response rather than to establish a universal ranking among the three training strategies.

Table 4 collects existing MC results. Comparisons are made within each row: different targets, domains, and marginals are not a common scale of system difficulty. The barrier formulation has the lowest reported errors in all four two-dimensional target cases and in the first Duffing (x_1, x_2) marginal. The latter values, 0.1055, 0.1030, and 0.1026, are close; their ordering alone does not demonstrate a substantial performance difference. The remaining

Table 4: Summary of the reported closed-loop MC L_1 density errors -

Case and parameter	Density compared	Soft constraint	Penalty	Barrier
SDOF target 1, $\omega = 1.18$	2D joint	0.1190	0.0780	**0.0693**
SDOF target 2, fixed target	2D joint	0.1950	0.0942	**0.0891**
Levitation target 1, $\omega = 5.0$	2D joint	0.4300	0.2820	**0.1100**
Levitation target 2, $\omega = 10.3$	2D joint	0.0952	0.0970	**0.0860**
Duffing target 1, $\omega = 1.8$	(x_1, x_2) marginal	0.1055	0.1030	**0.1026**
Duffing target 1, $\omega = 1.8$	(x_3, x_4) marginal	**0.0756**	0.0806	0.0830
Duffing target 2, $\omega = 1.0$	(x_1, x_2) marginal	0.1310	**0.1200**	0.1240
Duffing target 2, $\omega = 1.0$	(x_3, x_4) marginal	**0.0938**	0.0985	0.0995

Duffing marginals favor the soft-constraint or penalty formulation. These results support the barrier formulation's observed accuracy advantage in the two-dimensional cases while retaining the projection-dependent ranking in four dimensions.

5. Conclusion

The considered examples show that parameter-conditioned density and control networks can approximate the principal features of prescribed stationary targets, including annular and asymmetric densities and four-dimensional Duffing targets. The displayed density-network sweeps support reuse of the individual models at selected parameter values within their training ranges. They do not establish the same parameter-conditioned control accuracy for all three strategies, and the first Duffing sweep tests slice amplitude rather than the width along x_4. At representative parameters, closed-loop MC simulations reproduce the main target concentration patterns. The relative integrated density errors depend on the system and, for the Duffing examples, on the marginal being examined.

Under the adopted training and simulation settings, the barrier formulation offers an accuracy advantage over the soft-constraint and penalty formulations in the reported two-dimensional closed-loop density comparisons. As summarized in Table 4, it attains the lowest reported L_1 errors for both SDOF targets and both levitation targets. For the first levitation target, the error is 0.1100, compared with 0.4300 and 0.2820 for the soft-constraint and penalty formulations, respectively. This provides numerical evidence of closer target-density agreement in that case. The advantage is not uniform across all projections: the Duffing marginal rankings vary, and the first-target (x_1, x_2) errors are close.

Future work may examine full-joint and interval-wide validation, explicit physical input constraints, adaptive sampling, and computational scaling. Extension to time-varying targets would require a separate transient formulation and validation.

Acknowledgements

Wantao Jia was supported by the National Natural Science Foundation of China (11872306).

Declarations

Author contributions

Wantao Jia: Conceptualization, Methodology, Investigation, Software, Writing–review & editing, Supervision. **Xianzheng Li**: Writing–original draft, Visualization, Software, Methodology, Investigation. **Xiaotong Feng**: Writing–review & editing, Investigation, Formal analysis.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No experimental or observational datasets were used. The results reported in this article were generated through numerical training and simulations.

References

[1] Er G.-K. Probabilistic solutions of some multi-degree-of-freedom nonlinear stochastic dynamical systems excited by filtered Gaussian white noise. Computer Physics Communications 2014;185:4:1217–1222.

[2] Chen C, Chen N, Wu J.-L. CGNSDE: Conditional Gaussian neural stochastic differential equation for modeling complex systems and data assimilation. Computer Physics Communications 2024;304:109302.

[3] Åström K. J, Murray R. M, Feedback Systems: An Introduction for Scientists and Engineers, Princeton University Press, Princeton, 2010

[4] Kwakernaak H, Sivan R, Linear Optimal Control Systems, Wiley-Interscience, New York, 1972

[5] Soong T. T, Random Differential Equations in Science and Engineering, Academic Press, New York, 1973

[6] Lin Y. K, Cai G. Q, Probabilistic Structural Dynamics: Advanced Theory and Applications, McGraw-Hill, New York, 1995

[7] Wang H, Bounded Dynamic Stochastic Systems: Modelling and Control, Springer, London, 2000

[8] Wang H. Robust control of the output probability density functions for multivariable stochastic systems with guaranteed stability. IEEE Transactions on Automatic Control 1999;44:11:2103–2107.

[9] Guo L, Wang H. Generalized discrete-time PI control of output PDFs using square root B-spline expansion. Automatica 2005;41:1:159–162.

[10] Wang H, Zhang J. Bounded stochastic distributions control for pseudo-ARMAX stochastic systems. IEEE Transactions on Automatic Control 2001;46:3:486–490.

[11] Risken H, The Fokker–Planck Equation: Methods of Solution and Applications, 2nd Edition, Springer, Berlin, 1996

[12] Gardiner C. W, Stochastic Methods: A Handbook for the Natural and Social Sciences, 4th Edition, Springer, Berlin, 2009

[13] Annunziato M, Borzì A. A Fokker–Planck control framework for multidimensional stochastic processes. Journal of Computational and Applied Mathematics 2013;237:1:487–507.

[14] Zhu C. X, Zhu W. Q, Yang Y. F. Design of feedback control of a nonlinear stochastic system for targeting a pre-specified stationary probability distribution. Probabilistic Engineering Mechanics 2012;30:20–26.

[15] Zhu C. X, Zhu W. Q. Feedback control of nonlinear stochastic systems for targeting a specified stationary probability density. Automatica 2011;47:3:539–544.

[16] Yang Y, Wang Y, Huang Z. L. Probabilistic tracking control of dissipated Hamiltonian systems excited by Gaussian white noises. International Journal of Systems Science 2021;52:9:1790–1805.

[17] Jia W, Zhu W, Xu Y. Stochastic averaging of quasi-non-integrable Hamiltonian systems under combined Gaussian and Poisson white noise excitations. International Journal of Non-Linear Mechanics 2013;51:45–53.

[18] Jia W, Zhu W. Stochastic averaging of quasi-partially integrable Hamiltonian systems under combined Gaussian and Poisson white noise excitations. Nonlinear Dynamics 2014;76:2:1271–1289.

[19] Jia W, Jin Z, Ni F, Zhu W. Neural network-based probabilistic tracking control for levitation systems under stochastic track irregularities. Probabilistic Engineering Mechanics 2025;82:103866.

[20] Jia W, Jin Z, Zan W, Ni F. Design of feedback control of levitation system excited by track irregularity for targeting a pre-specified stationary probability density. Journal of Sound and Vibration 2025;607:119071.

[21] Zhu W. Q, Nonlinear Stochastic Dynamics and Control in Hamiltonian Formulation, Science Press, Beijing, 2003

[22] Zhu W. Q. Stochastic averaging methods in random vibration. ASME Applied Mechanics Reviews 1988;41:5:189–199.

[23] Roberts J. B, Spanos P. D. Stochastic averaging: an approximate method of solving random vibration problems. International Journal of Non-Linear Mechanics 1986;21:2:111–134.

[24] Kumar P, Narayanan S. Solution of the Fokker–Planck equation by finite element and finite difference methods for nonlinear systems. Sadhana 2006;31:4:445–461.

[25] Masud A, Bergman L. A, Solution of the four dimensional Fokker–Planck equation: still a challenge, in: Proceedings of the ICOSSAR 2005, Rome, Italy, 2005

[26] Annunziato M, Borzì A. Optimal control of probability density functions of stochastic processes. Mathematical Modelling and Analysis 2010;15:4:393–407.

[27] Sun J. Q, Stochastic Dynamics and Control, Elsevier, Amsterdam, 2006

[28] Xu Y, Zan W, Jia W, Kurths J. Path integral solutions of the governing equation of SDEs excited by Lévy white noise. Journal of Computational Physics 2019;394:41–55.

[29] Zan W, Jia W, Xu Y. Reliability of dynamical systems with combined Gaussian and Poisson white noise via path integral method. Probabilistic Engineering Mechanics 2022;68:103252.

[30] Zan W, Xu Y, Jia W, Kurths J. Response statistics of single-degree-of-freedom systems with Lévy noise by improved path integral method. Communications in Nonlinear Science and Numerical Simulation 2023;120:107144.

[31] Kappen H. J. Path integrals and symmetry breaking for optimal control theory. Journal of Statistical Mechanics: Theory and Experiment 2005;2005:11:P11011.

[32] Xu Y, Zhang H, Li Y, Zhou K, Liu Q, Kurths J. Solving Fokker–Planck equation using deep learning. Chaos 2020;30:1:013133.

[33] Zhai J, Dobson M, Li Y, A deep learning method for solving Fokker–Planck equations, in: Bruna J, Hesthaven J, Zdeborova L (Eds.), Proceedings of the 2nd Mathematical and Scientific Machine Learning Conference, Vol, 145 of Proceedings of Machine Learning Research, PMLR, Virtual Conference, 2022;pp.568–597

[34] Zhang D, Lu L, Guo L, Karniadakis G. E. Quantifying total uncertainty in physics-informed neural networks for solving forward and inverse stochastic problems. Journal of Computational Physics 2019;397:108850.

[35] Chen X, Yang L, Duan J, Karniadakis G. E. Solving inverse stochastic problems from discrete particle observations using the Fokker–Planck equation and physics-informed neural networks. SIAM Journal on Scientific Computing 2021;43:3:B811–B830.

[36] Raissi M, Perdikaris P, Karniadakis G. E. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational Physics 2019;378:686–707.

[37] Halder R, Stabile G, Rozza G. Coupling physics informed neural networks with external solvers. Computer Physics Communications 2027;330:110424.

[38] Yin H, Liu P, Ng B. F. Accelerating Acoustic Agglomeration Predictions: A Physics-informed Neural Network with Fourier Featuring for Sound Pressure Gradient and Stokes Drag. Computer Physics Communications 2026;:110451.

[39] Feng X, Zan W, Ma S, Jia W. Deep neural network-Monte Carlo framework for stochastic dynamic response prediction under combined continuous and jump excitations. Communications in Nonlinear Science and Numerical Simulation 2026;163:110833.

[40] Thakur A, Zahr M. J. Feature tracking in physics-informed neural networks via joint optimization of analytical nonlinear manifolds: Application to boundary layers. Computer Physics Communications 2027;330:110419.

[41] Liu Y, Cai L, Chen Y, Liu Q. Fluid–structure interaction training of integral conservation physics-informed neural networks for blood flow in elastic vessels. Computer Physics Communications 2026;324:110131.

[42] Mowlavi S, Nabi S. Optimal control of PDEs using physics-informed neural networks. Journal of Computational Physics 2023;473:111731.

[43] Lu L, Pestourie R, Yao W, Wang Z, Verdugo F, Johnson S. G. Physics-informed neural networks with hard constraints for inverse design. SIAM Journal on Scientific Computing 2021;43:6:B1105–B1132.

[44] Norman G, Wentz J, Kolla H, Maute K, Doostan A. Constrained or unconstrained? neural-network-based equation discovery from data. Computer Methods in Applied Mechanics and Engineering 2025;436:117684.

[45] Jia W, Jiao Z, Jin Z. Neural controller for targeting a desired stationary distribution in stochastic systems. International Journal of Non-Linear Mechanics 2025;174:105058.

[46] Jia W, Hu M, Zan W, Ni F. Data-driven methods for the inverse problem of suspension system excited by jump and diffusion stochastic track excitation. International Journal of Non-Linear Mechanics 2024;166:104819.